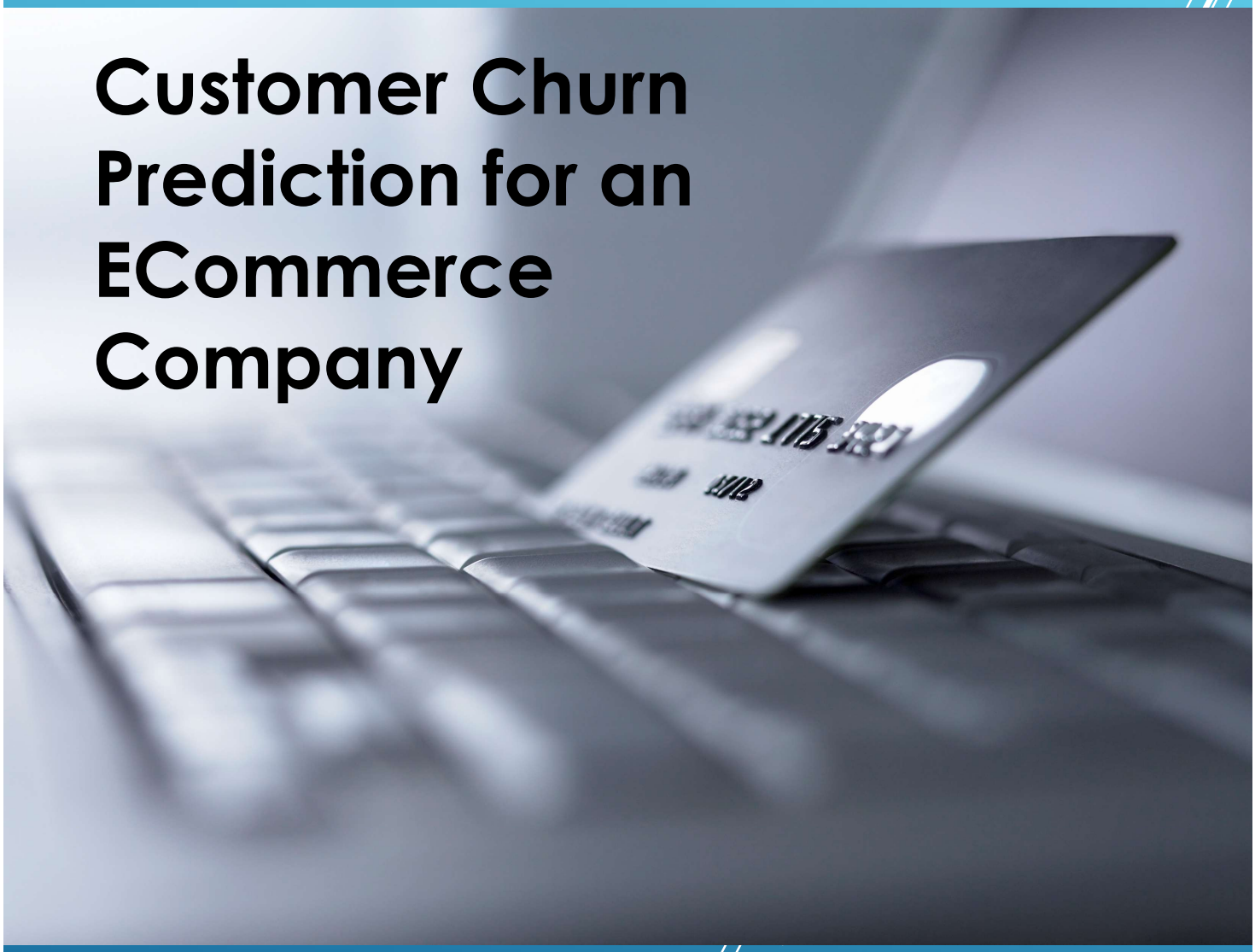
Customer Churn Prediction for an ECommerce Company



Capstone Project

Business Report

by

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Literature Review

- Customer churn prediction is crucial for ecommerce industries because of the highly competitive nature of the market
- The number of e-commerce businesses has increased as global e-commerce sales have more than doubled since 2017
- This literature review explores key research findings, methodologies and factors influencing customer churn in e-commerce sector

Definition and Importance of Churn Prediction

- Customer churn is defined as the loss of customers or the cessation of their transactions with a business
- Churn prediction models aim to identify customers who are likely to stop using a service, allowing companies to implement preemptive retention strategies
- According to Reichheld and Sasser (1990), a 5% increase in customer retention can lead to an increase in profits ranging from 25% to sometimes even 95%, highlighting the economic significance of churn prediction

Customer Retention Marketing Stats

- Acquiring a new customer can cost four to five times more than retaining an existing customer
- Increasing customer retention by 5% can increase profits from 25-95%
- The success rate of selling to a customer you already have is 60-70%, while the success rate of selling to a new customer is 5-20%
- One customer experience agency found loyal customers are 5 times as likely to repurchase, 5 times as likely to forgive, 4 times as likely to refer, and 7 times as likely to try a new offering
- U.S. companies lose $136.8 billion per year due to avoidable consumer switching
- American Express found 33% of customers will consider switching companies after just one instance of poor customer service

Sources:
https://www.outboundengine.com/blog/customer-retention-marketing-vs-customer-acquisition-marketing/
https://hbr.org/2014/10/the-value-of-keeping-the-right-customers
https://www.forbes.com/councils/forbesbusinesscouncil/2022/12/12/customer-retention-versus-customer-acquisition/

Introduction

- Customer churn is a critical challenge faced by e-commerce companies, as losing customers can significantly impact revenue and growth
- In an increasingly competitive market, retaining existing customers is often more cost-effective than acquiring new ones
- Therefore, understanding the factors that contribute to customer churn and being able to predict which customers are at risk of leaving can provide e-commerce companies with valuable insights to develop targeted retention strategies
- This project aims to build a predictive model that identifies customers who are likely to churn, enabling the company to take proactive measures to retain them
- By analyzing historical customer data, including demographics, purchasing behavior, engagement metrics, and other relevant features, the model can detect patterns and signals indicative of potential churn

Need of the Project

- Minimize Customer Churn
  - Develop strategies to reduce the number of customers leaving the ecommerce site
- Identify At-Risk Customers
  - Analyze customer behavior to pinpoint those most likely to churn
  - Target them with personalized promotional offers to improve retention
- Optimize Promotional Expenses
  - Identify spending patterns of loyal customers to efficiently allocate promotional resources
  - Reduce unnecessary costs

Business Opportunity

- Improved Customer Retention and Loyalty
  - Focusing on minimizing churn and optimizing promotional efforts can lead to a stronger, more loyal customer base
  - This enhances the brand's reputation and creates a competitive advantage in the market
- Increased Profitability
  - By retaining customers and optimizing spending, the business can increase its profitability, ensuring long-term sustainability and growth
- Data-Driven Decision Making
  - Leveraging customer data to inform retention and promotional strategies can improve decision-making across the organization
  - This will lead to more effective business strategies

Problem Statement

An E Commerce company is facing a lot of competition in the current market, and it has become a challenge to retain the existing customers in the current situation.

Hence, the company wants to develop a model through which they can do churn prediction of the accounts and provide segmented offers to the potential churners. In this company, account churn is a major thing because 1 account can have multiple customers. hence by losing one account the company might be losing more than one customer.

The goal is to develop a churn prediction model for this company and provide business recommendations on the campaign. The campaign suggestion should be unique and be very clear on the campaign offer because the recommendation needs to be approved by the revenue assurance team whose goal is to minimize the loss to the company.

Data Report

Shape of the Data

- There are 11260 rows in the dataset
- There are 19 columns in the dataset

Frequency of the Data

- Looking at the column rev_growth_yoy, we can see that the data has been collected for at least 2 years or more
- Data has been collected from
  - the registration details of the customer (gender, city tier, etc.)
  - transactions made by the customer (monthly revenue, payment mode, etc.)
  - Interaction of the customer with the customer service (complain_l12m, service_score, etc.)

Data Dictionary

Column Name	Description	Data Type
Accountid	Account unique identifier - continuous	Integer
Churn	Account churn flat (Yes – 1, No – 0)- categorical	Integer
Tenure	Tenure of the account **(in months)** - continuous	Integer
City_tier	Tier of the city (values 1 to 3) categorical	Integer
Cc_contacted_ly	Number of times customer care was called by the account in the last 12 months - continuous	Integer
Payment	Preferred payment mode categorical	String
Gender	Gender of the primary account holder- categorical	String
Service_score	Satisfaction score given by customers of the account on service provided by company (0 to 5) - categorical	Integer
Account_user_count	Number of customers tagged with the account - categorical	Integer
Account_segment	Account segmentation of the basis of spend (Regular, Regular Plus, Super, Super Plus, HNI)- categorical	String
Cc_agent_score	Satisfaction score given by customers of the account on customer care service provided by company (1 to 3) - categorical	Integer
Marital_status	Marital status of the primary customer of the account (Marries, Single, Divorced) - categorical	String
Rev_per_month	Monthly average revenue generated by account in last 12 months - continuous	Float
Complain_ly	Any complaints has been raised by account in last 12 months (No - 0 or Yes - 1) categorical	Integer
Rev_growth_yoy	revenue growth percentage of the account (last 12 months vs last 24 to 13 month)- continuous	Float
Coupon_used_for_payment	How many times customers have used coupons to do the payment in last 12 months - continuous	Integer
Day_since_cc_connect	Number of days since no customers in the account has contacted the customer care- continuous	integer
Cashback	Monthly average cashback generated by account in last 12 months - continuous	float
Login_device	Preferred login device of the customers in the account (Computer/Mobile) - categorical	string

Table 1

Fixing Data Entry Errors

- Tenure has some invalid values “#” which have been changed to null
- Gender has some values “M” and “F” which were changed to “Male” and “Female” respectively
- Account user count has some invalid values “@” which have been changed to null
- Account Segment has some values “Super +” and “Regular +” which have been changed to “Super Plus” and “Regular Plus”
- Revenue growth field, day since cc connect and cashback have some invalid values “$” which have been changed to null
- Coupon field has some invalid values “#’, “$” and “*” which have been changed to null
- Login device has some invalid values “&&&&” which have been changed to null

Statistical Summary – Continuous Variables

	count	mean	std	min	25%	50%	75%	max
tenure	11042.0	11.03	12.88	0.0	2.00	9.00	16.00	99.0
cc_contacted_ly	11158.0	17.87	8.85	4.0	11.00	16.00	23.00	132.0
rev_per_month	10469.0	6.36	11.91	1.0	3.00	5.00	7.00	140.0
rev_growth_yoy	11257.0	16.19	3.76	4.0	13.00	15.00	19.00	28.0
coupon_used_for_payment	11257.0	1.79	1.97	0.0	1.00	1.00	2.00	16.0
day_since_cc_connect	10902.0	4.63	3.70	0.0	2.00	3.00	8.00	47.0
cashback	10787.0	196.24	178.66	0.0	147.21	165.25	200.01	1997.0

Fig 1

- Mean is higher than the median for all the continuous variables suggesting the distribution is positively skewed
- Maximum values for most of the variables are much higher than 75% of their respective values suggesting the presence of outliers

Heatmap of Continuous Variables

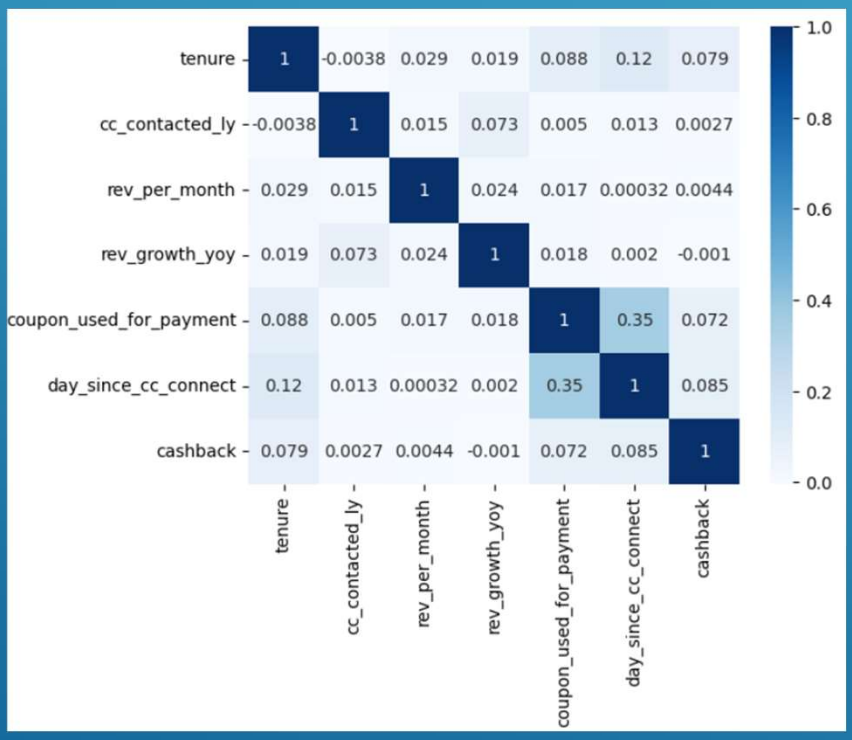


Fig 2

- Weak correlation between coupon used for payment and day since cc connect
- Not much correlation between any other variables

Statistical Summary – Categorical Variables

	count	unique	top	freq
churn	11260	2	0	9364
city_tier	11148.0	3.0	1.0	7263.0
payment	11151	5	Debit Card	4587
gender	11152	2	Male	6704
service_score	11162.0	6.0	3.0	5490.0
account_user_count	10816.0	6.0	4.0	4569.0
account_segment	11163	5	Regular Plus	4124
cc_agent_score	11144.0	5.0	3.0	3360.0
marital_status	11048	3	Married	5860
complain_ly	10903.0	2.0	0.0	7792.0
login_device	10500	2	Mobile	7482

Fig 3

- 17% of the customers have churned
- 65% of the customers are from city tier 1
- 41% of the customers pay by debit card
- 60% of the customers are male
- 49% of the customers have rated the service as moderate(3)
- 41% of the customers have 4 user accounts
- 37% of the accounts are regular plus accounts
- 30% of the customers have given a moderate rating (3) for the customer service agent
- 52% of the customers are married
- Nearly 30% of the customers have complained last year
- 66% of the customers use their mobiles to make their transactions

Exploratory Data Analysis - Univariate

Continuous Variables

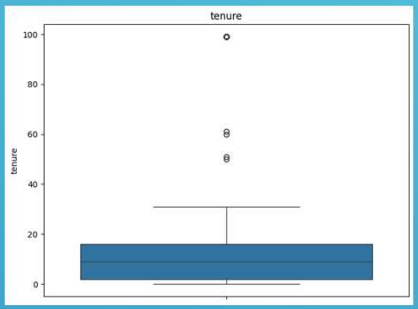


Fig 4

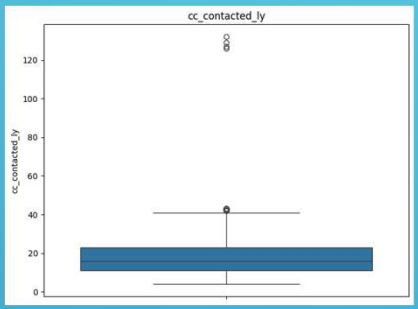


Fig 5

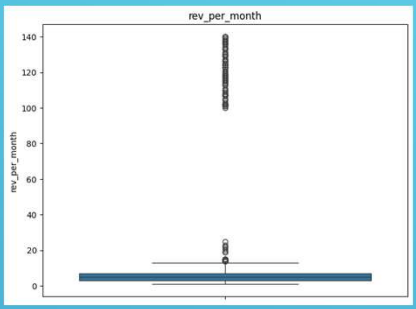
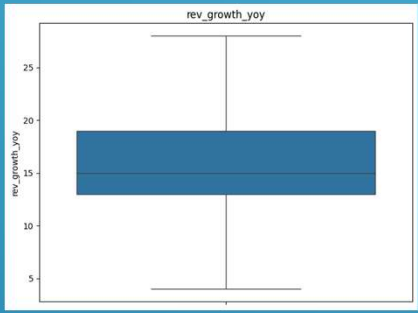


Fig 6



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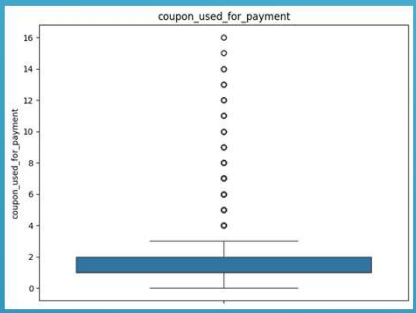

Fig 8

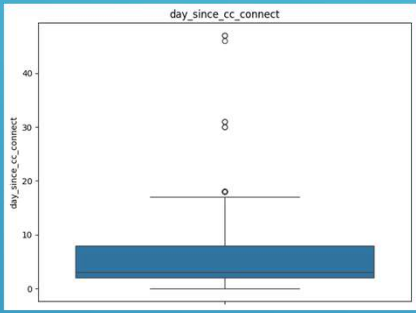

Fig 9

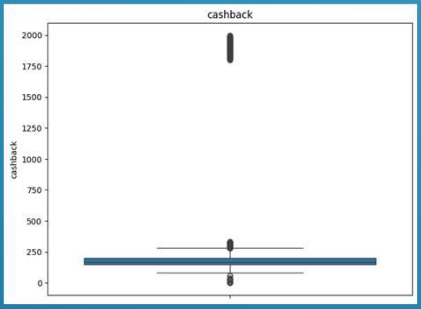

Fig 10

- Outliers in all the continuous variables except revenue growth year over year
- Most customers have contacted customer service once every 5 days
- Most accounts have cashback of 250 dollars or less

Exploratory Data Analysis - Univariate

Categorical Variables

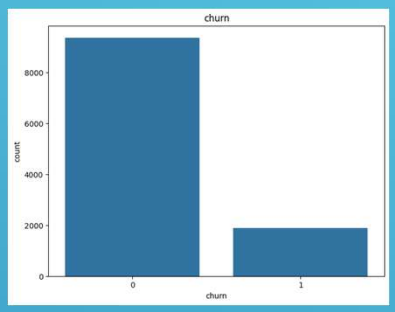

Fig 11

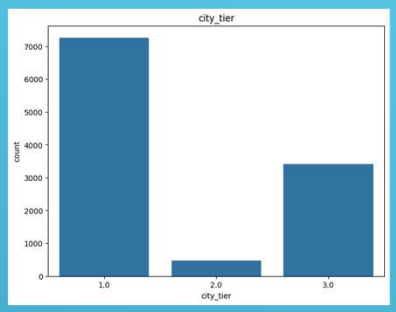

Fig 12

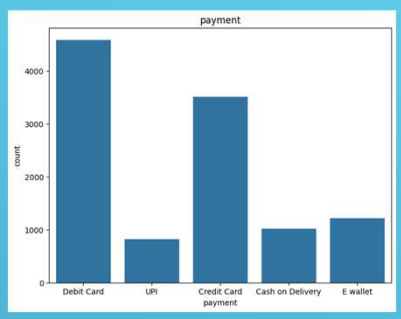

Fig 13

- Significantly more churners than non churners suggesting data imbalance
- City tier 1 has highest number of customers, city tier 2 has the least
- Most customers pay using debit card and credit card, UPI is used very less

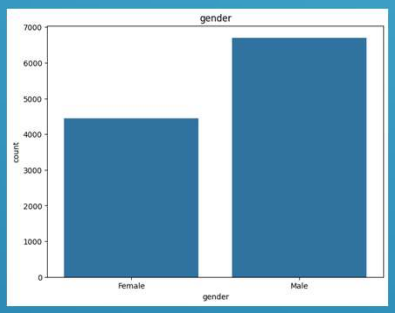

Fig 14

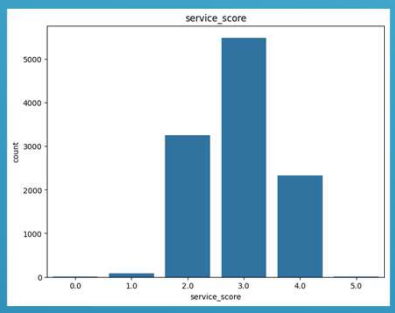

Fig 15

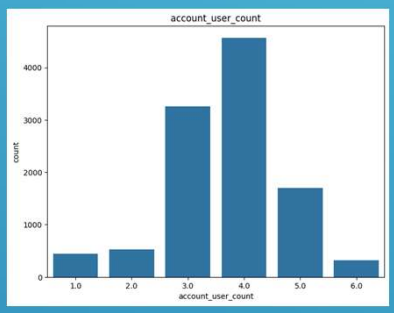

Fig 16

- More male customers than female customers
- Most customers have rated the service 2 to 4
- Most accounts have 4 users followed by 3

Exploratory Data Analysis - Univariate

Categorical Variables

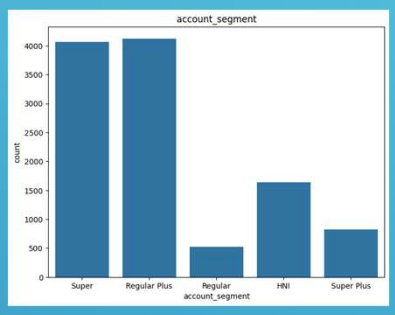

Fig 17

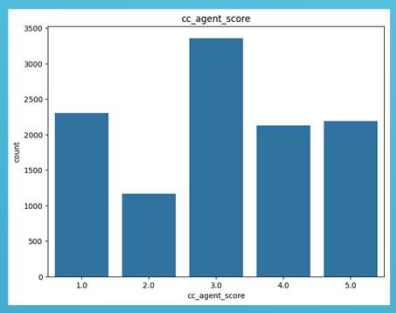

Fig 18

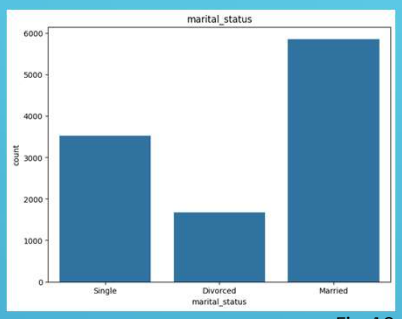

Fig 19

- Most customers come under regular plus and super account segments
- Most customers have given a moderate rating to the customer service agent
- Most customers are married

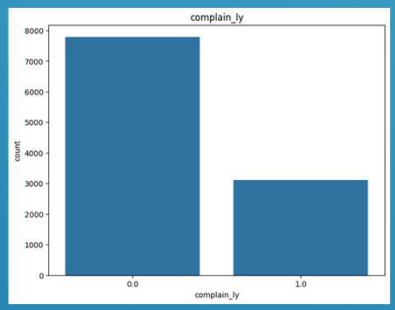

Fig 20

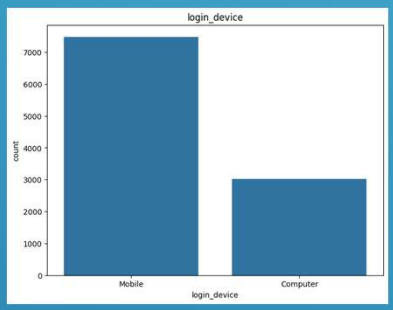

Fig 21

- Roughly 25% customers have complained last year
- Most of the customers use mobile to do their purchases

Exploratory Data Analysis - Bivariate

Categorical Variables vs Churn

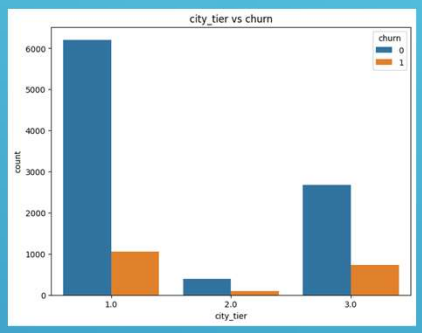

Fig 22

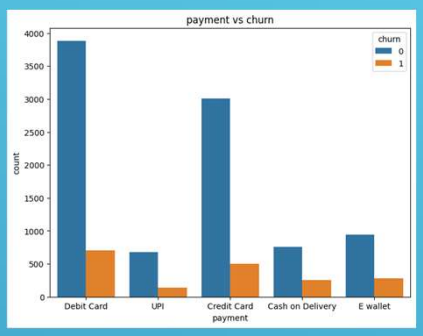

Fig 23

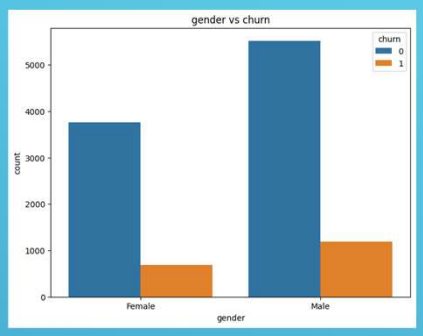

Fig 24

- City tier 3 customers have churned in higher proportion
- Credit card users have churned in higher proportion
- Males have a higher churn rate than females

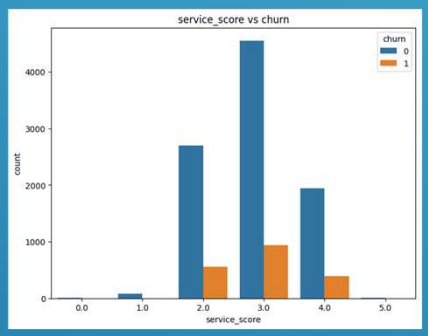

Fig 25

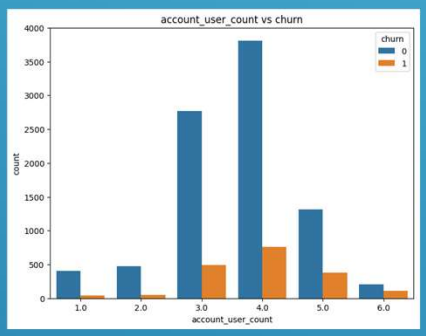

Fig 26

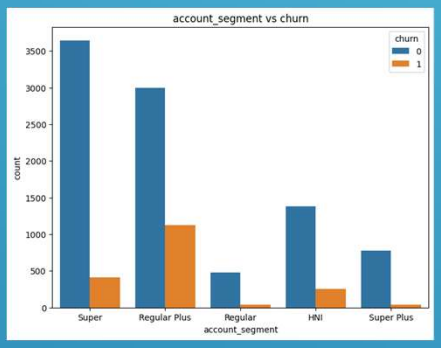

Fig 27

- Churners have rated the service 2 to 4
- Customers with 3 user accounts have churned at a high proportion
- Churn rate is high for regular plus customers compared to super category

Exploratory Data Analysis - Bivariate

Categorical Variables vs Churn

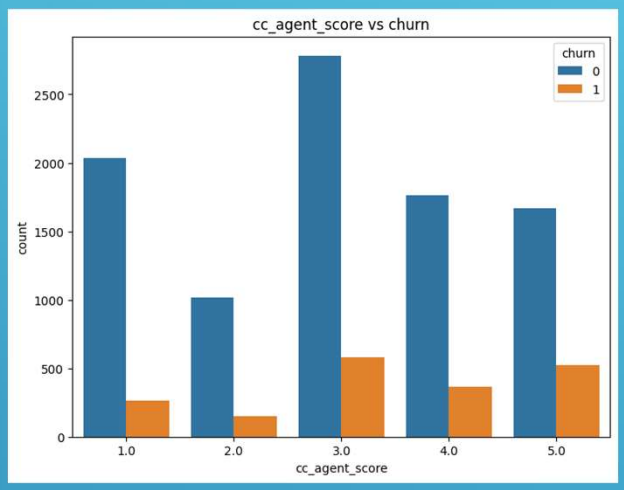

Fig 29

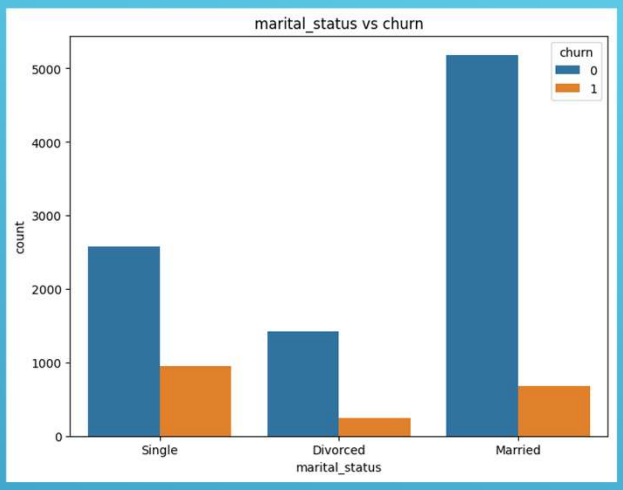

Fig 30

- Most churners have given moderate to good rating to the customer service agent
- Churn rate is high for single customers

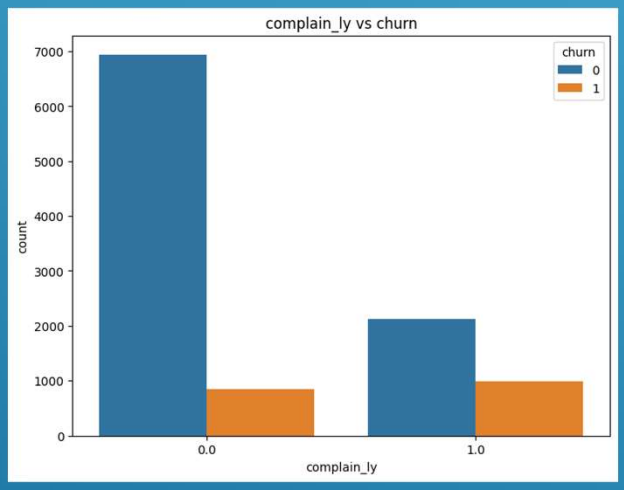

Fig 30

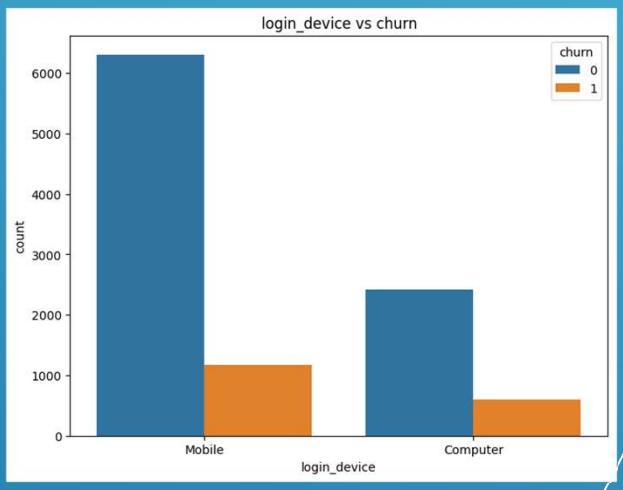

Fig 31

- Churners have rated the service 2 to 4
- Customers with 3 user accounts have churned at a high proportion
- Churn rate is high for regular plus customers compared to super category

Exploratory Data Analysis - Bivariate

Continuous Variables vs Churn

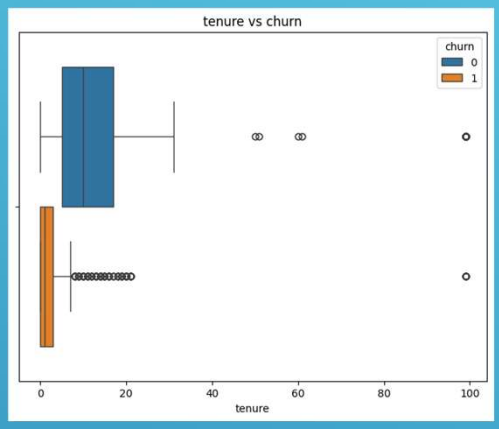

Fig 32

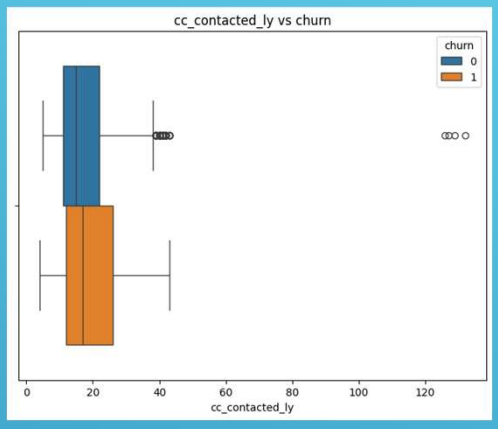

Fig 33

- Churners have less tenure meaning
- Churners have contacted the customer service more in the last 12 months

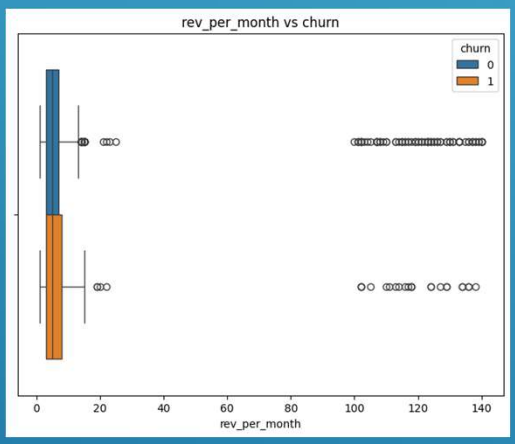

Fig 34

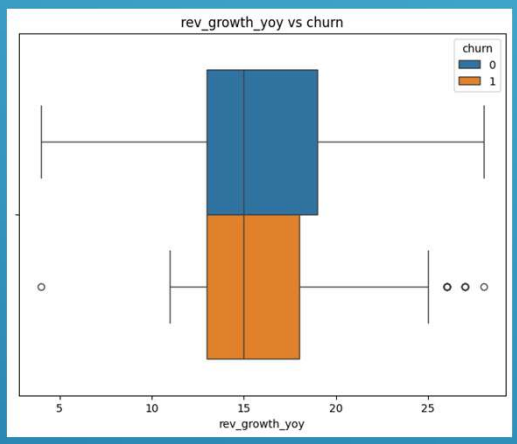

Fig 35

- Revenue per month is almost the same for churners and non-churners
- Revenue growth year over year is comparatively less for churners

Exploratory Data Analysis - Bivariate

Continuous Variables vs Churn

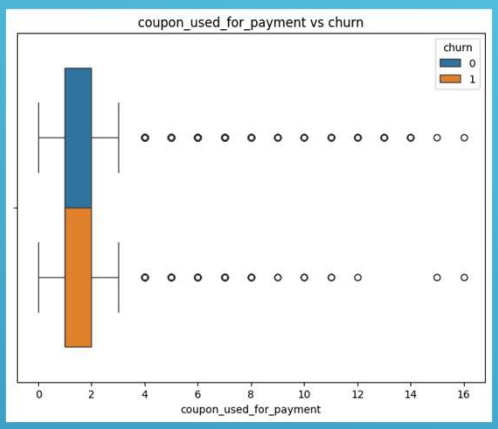

Fig 36

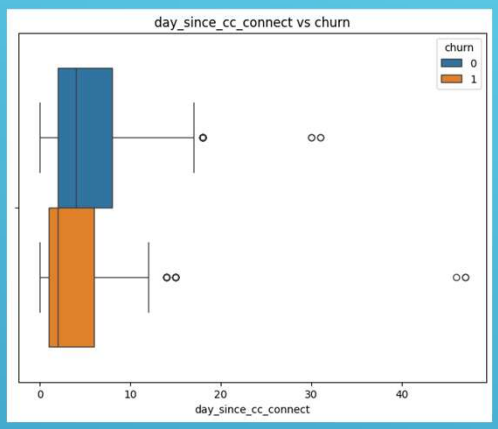

Fig 37

- Coupon usage has no impact on churning
- More customers have churned recently

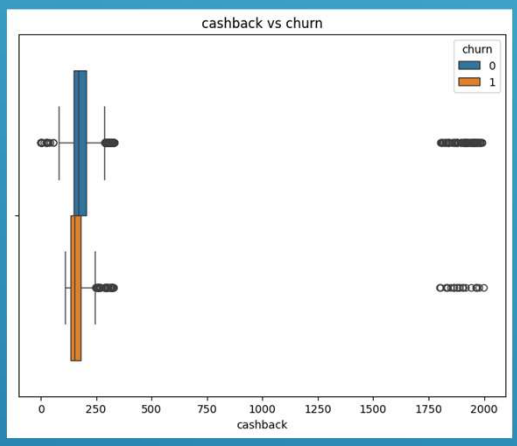

Fig 38

- Churners have slightly less cashback than non-churners

Exploratory Data Analysis - Multivariate

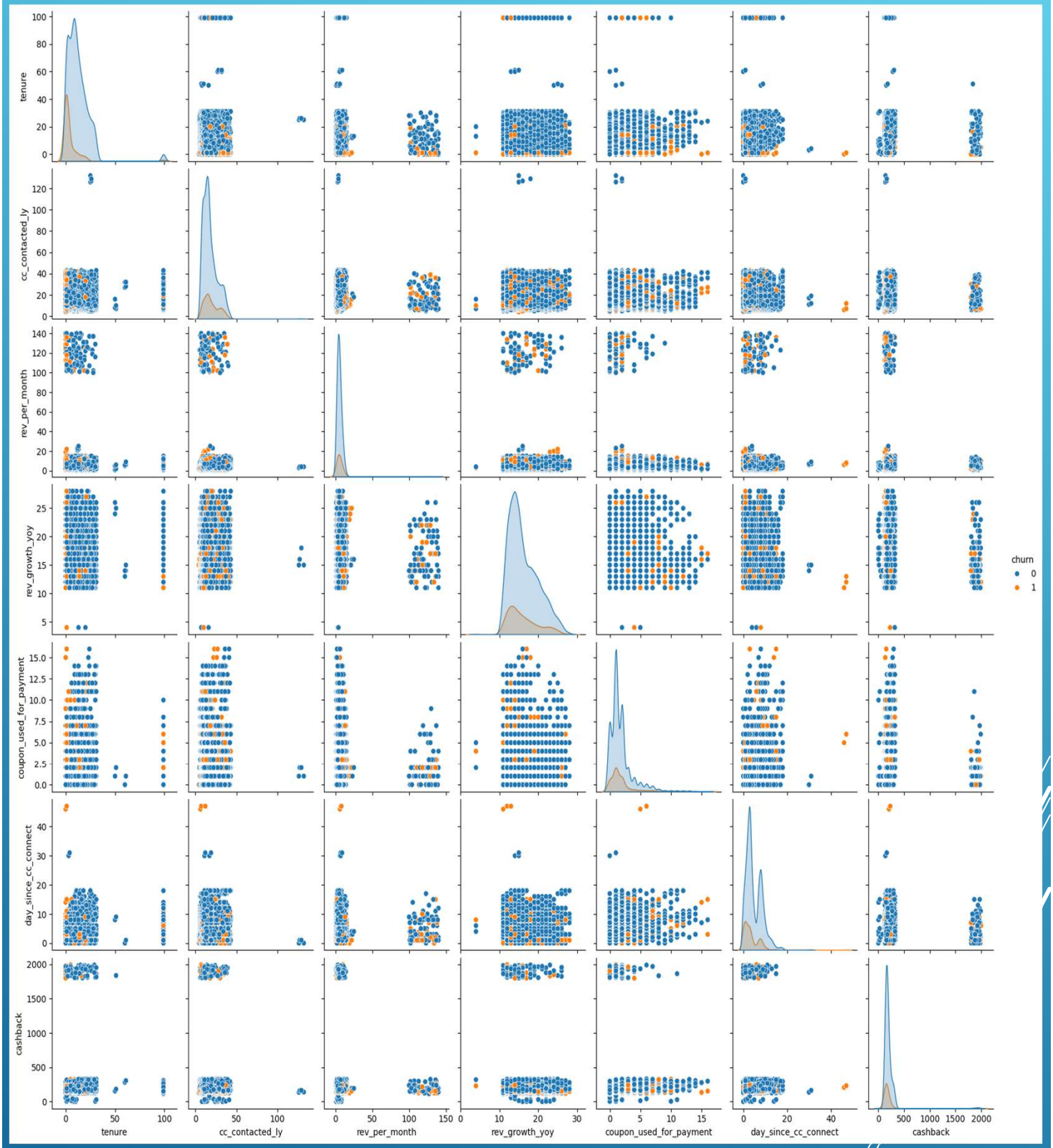

Fig 39

- No patterns are clearly evident indicating lack of correlation between continuous variables

Customer Segmentation

- Customers were grouped into different clusters using K-means Clustering

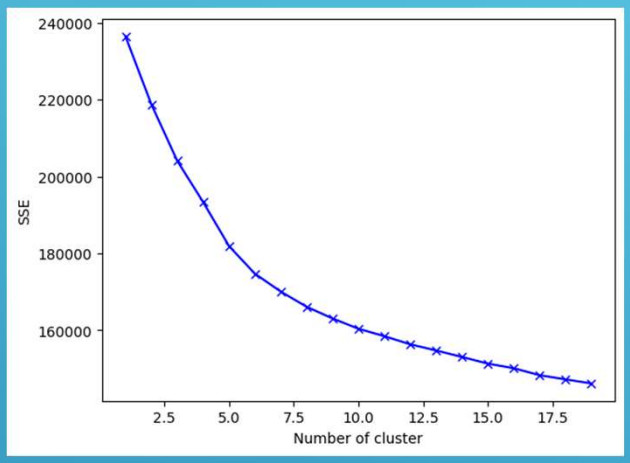

Fig 40

- Five clusters separate the features sharply

Count Plot Clustered Data with Churn

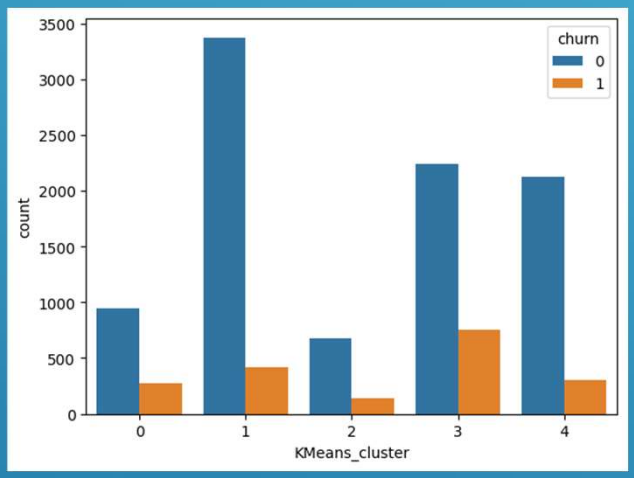

Fig 41

- Cluster 3 has churned the most

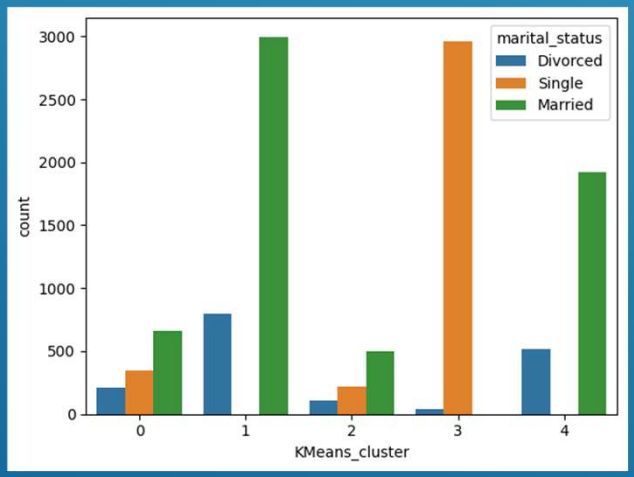

Fig 42

- Cluster 3 has the highest number of single customers

Actionable Insights from EDA

- 17% of the customers have churned which is higher than the industry standard of 9 to 10%
- Market share in city tier 2 and 3 is very less and can be improved
- Targeted marketing to get more female customers
- Service can be improved as most customers have given only a moderate feedback
- 41% of the customers have 4 user accounts
- Targeted marketing to attract more customers in the Super plus and HNI segments
- 25% of the customers have complained last year which needs to be addressed
- 66% of the customers use their mobiles to make their transactions
- Churn rate is high for regular plus customers compared to super category which needs to be analyzed further
- More cashback offers for churners can be given

Data Cleaning and Preprocessing

- Removal of Unwanted Variable
    - Account id field is the unique identifier of a row was removed from the dataset because it won't add any value to the model predicting the churn
- Handling Missing Values
    - There are null values in all the fields except churn field
    - Imputing these null values ensures all available data points are utilized
    - Most machine learning algorithms cannot handle missing values
    - If null values are simply dropped, it may introduce bias to the model
    - Imputing minimizes the bias by providing a reasonable estimation to the missing values
    - **The continuous features' missing values are imputed with median because the data is positively skewed and the median is a better measure than the mean**
    - **The categorical missing values are imputed with their respective modes (most frequently occurring value) which is a reasonable assumption for filling gaps**
- Duplicate Value Check
    - There are no duplicates in the dataset
- Outlier Treatment
    - There are outliers in the dataset
    - It can be treated using inter quartile range (IQR) method, clipping the values to
    - Q1 - 1.5 times IQR as the lower limit
    - Q3 + 1.5 times IQR as the upper limit
    - Outliers need to be handled carefully because they are still valid data
    - It is better to develop models with and without treating the outliers and compare their performance
    - Domain experts can be consulted with to see what can be done with the outliers

Data Cleaning and Preprocessing

Variable Transformation

Data Encoding

- Machine learning algorithms can only work with numerical or boolean values
- Hence, all the categorical values need to be encoded
- Variables to be encoded:
  - Payment
  - Gender
  - Account segment
  - Marital status
  - Login device
- Account segment has been encoded ordinally:
  - Regular is encoded as 1
  - Regular plus is encoded as 2
  - Super is encoded as 3
  - Super plus is encoded as 4
  - HNI is encoded as 5
- Dummy encoding has been done for all other variables

Scaling of the Data

- The data has been scaled using Standard Scaler so that equal weightage is given to all the features by the machine learning algorithm
- After scaling, mean of all the variables is close to 0 and the standard deviation of all the variables is close to 1

Statistical Summary after Scaling

	count	mean	std	min	25%	50%	75%	max
tenure	11260.0	0.0	1.0	-1.1533	-0.9283	-0.1408	0.6467	3.0093
cc_contacted_ly	11260.0	-0.0	1.0	-1.5713	-0.7772	-0.2099	0.5842	12.9503
rev_per_month	11260.0	0.0	1.0	-1.4762	-0.7817	-0.0871	0.6075	2.6912
rev_growth_yoy	11260.0	-0.0	1.0	-3.2453	-0.8499	-0.3176	0.7471	3.1426
coupon_used_for_payment	11260.0	-0.0	1.0	-1.3385	-0.4312	-0.4312	0.4760	1.8370
day_since_cc_connect	11260.0	0.0	1.0	-1.3014	-0.7289	-0.4426	0.7024	2.8493
cashback	11260.0	-0.0	1.0	-2.3760	-0.6746	-0.2762	0.4596	2.1610

Fig 43

Heatmap after Scaling

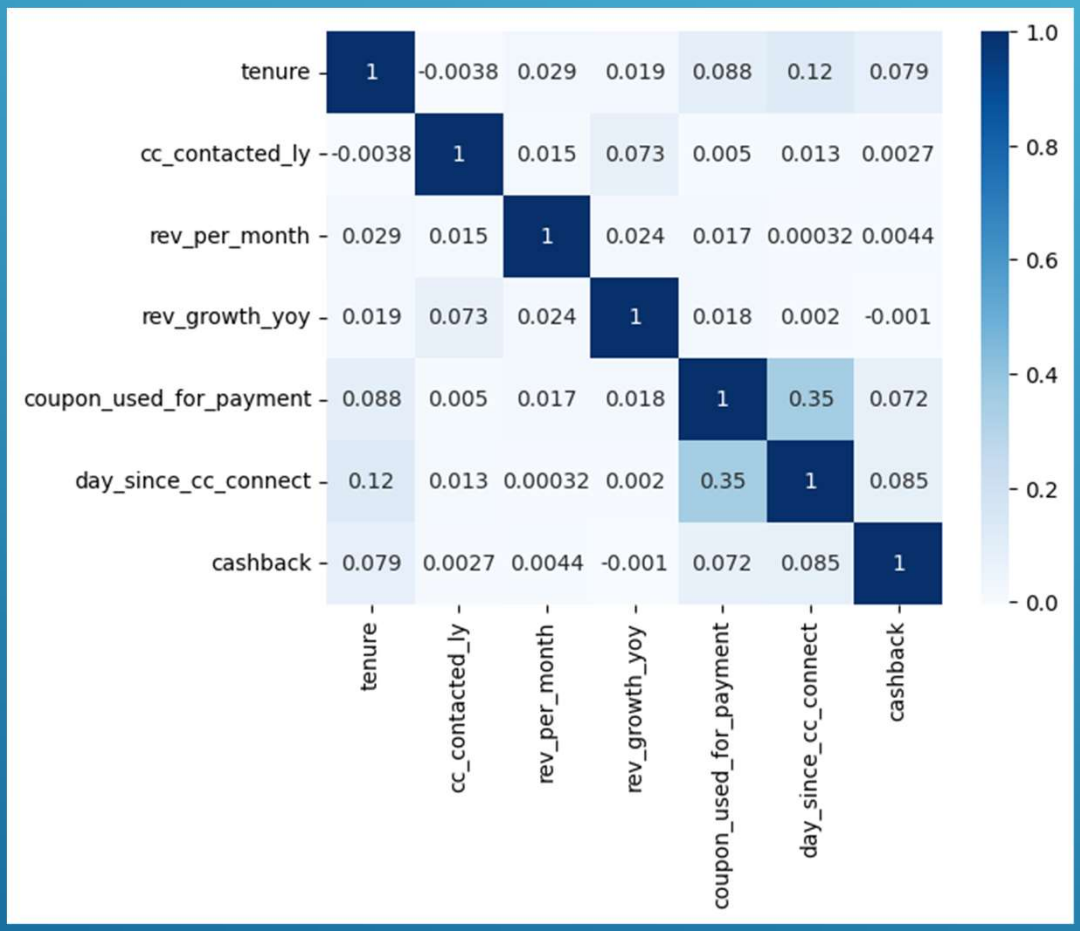

Fig 44

Train Test Split

- The data is split into training data and test data
- 70% of the data is kept for training
- 30% of the data is kept for testing
  - X_train – training data dependent variables (7882 rows and 22 columns)
  - X_test – test data for dependent variables (3378 rows and 22 columns)
  - y_train – training data target variable (churn) (7882 rows and 1 column)
  - y_test – test data target variable (churn) (3378 rows and 1 column)
- All the models are fit on the training data
- Then, the test data is used for predictions and comparison

Model Building

- Predicting whether a customer will churn (or not) is a classification problem
- The following models were built and analyzed:
  - Logistic Regression
  - K Nearest Neighbors (KNN)
  - Linear Discriminant Analysis (LDA)
  - Gaussian Naïve Bayes
  - Decision Tree
  - Random Forest
  - Bagging Classifier
  - AdaBoost Classifier
  - Gradient Boost Classifier
  - XG Boost Classifier

Model Metrics for the Classification Problem

- Accuracy = Number of Correct Predictions / Total number of predictions
- Precision = Number of True Positive Predictions/ Total Number of Positive Predictions
- Recall = Number of True Positive Predictions/(Number of True Positive Predictions + Number of False Negative Predictions)
- F1 Score(Harmonic mean of Precision and Recall)
  = 2 X (Precision X Recall)/(Precision + Recall)
- AUC – ROC Curve:
  ROC - Computes the False Positive Rate (FPR) and True Positive Rate (TPR) at different thresholds
  AUC - Computes the Area Under the ROC Curve (AUC score).

Wrongly identifying a churner as a non-churner or vice versa will be a great loss of revenue to the company. Hence, "Recall" is the most important metric for this business problem.

Model 1 – Logistic Regression

- Logistic regression is a statistical model used for binary classification tasks
- It estimates the probability that a given input belongs to a certain class by fitting a logistic function (S-shaped curve) to the data
- The output is a value between 0 and 1
- It can be interpreted as:
  - If the output is greater than or equal to 0.5 (threshold), the customer will churn
  - If the output is less than 0.5, the customer will not churn

Model Performance

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                  churn   No. Observations:                 7882
Model:                          Logit   Df Residuals:                     7860
Method:                           MLE   Df Model:                           21
Date:                Sat, 14 Sep 2024   Pseudo R-squ.:                  0.3309
Time:                        10:23:03   Log-Likelihood:                -2394.7
converged:                       True   LL-Null:                       -3579.1
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     -2.6192      0.060    -43.590      0.000      -2.737      -2.501
tenure                    -1.6280      0.069    -23.649      0.000      -1.763      -1.493
city_tier                  0.2124      0.042      5.087      0.000       0.131       0.294
cc_contacted_ly            0.2114      0.038      5.629      0.000       0.138       0.285
service_score             -0.1082      0.042     -2.596      0.009      -0.190      -0.026
account_user_count         0.3282      0.041      7.952      0.000       0.247       0.409
account_segment           -0.0826      0.053     -1.548      0.122      -0.187       0.022
cc_agent_score             0.3655      0.038      9.557      0.000       0.291       0.440
rev_per_month              0.3714      0.038      9.882      0.000       0.298       0.445
complain_ly                0.7270      0.035     20.887      0.000       0.659       0.795
rev_growth_yoy            -0.1037      0.039     -2.693      0.007      -0.179      -0.028
coupon_used_for_payment    0.2199      0.045      4.937      0.000       0.133       0.307
day_since_cc_connect      -0.3708      0.049     -7.594      0.000      -0.466      -0.275
cashback                  -0.0044      0.063     -0.070      0.945      -0.128       0.119
payment_Credit Card       -0.3140      0.061     -5.174      0.000      -0.433      -0.195
payment_Debit Card        -0.3068      0.062     -4.922      0.000      -0.429      -0.185
payment_E wallet          -0.0662      0.051     -1.285      0.199      -0.167       0.035
payment_UPI               -0.1794      0.045     -3.984      0.000      -0.268      -0.091
gender_Male                0.1498      0.038      3.963      0.000       0.076       0.224
marital_status_Married    -0.0698      0.057     -1.233      0.218      -0.181       0.041
marital_status_Single      0.3586      0.053      6.710      0.000       0.254       0.463
login_device_Mobile       -0.1924      0.036     -5.371      0.000      -0.263      -0.122
==========================================================================================
```

Fig 45

- Tenure has a coefficient of -1.62
- Complain last year has a coefficient of 0.72
- The sigmoid function σ(z) is used to map the log-odds to a probability.
- It is defined as:
- σ(z)=1/(1+e^{-z}) where z is the log-odds value
- **Change in Log-Odds**: The log-odds of churn decrease by 1.62 for every unit increase of tenure
- If initial probability of churn is 0.5 after a unit increase of tenure the probability reduces to (1/(1+$e^{-(-1.62)}$)) = 0.165 (16.5)
- Confusion Matrix, Classification Report and ROC-AUC Curves are provided in Appendix

Model 2 – K Nearest Neighbors (KNN)

- K-Nearest Neighbors (KNN) is a simple, non-parametric, and lazy learning algorithm used in machine learning
- Each data point is represented as a point in a multi-dimensional space, where each dimension corresponds to a feature of the data
- When a new data point needs to be classified, the algorithm calculates the distance between this point and all other points in the dataset
- The most common distance metric used is the Euclidean distance, though other metrics like Manhattan or Minkowski distance can also be used
- The algorithm identifies the 'k' nearest neighbors to the new data point. The value of 'k' is a user-defined constant, which represents the number of neighbors to consider
- The algorithm checks the labels of these 'k' neighbors and assigns the most common label (i.e., the majority vote) to the new data point

Model Performance (Classification Report given in Appendix)

- Recall for non-churners for test data is 98%
- Recall for churners for test data is 87% which Is 5% less than training data indicating some overfitting
- KNN is performing much better than Logistic Regression
- Area under the curve is close to 1 which implies the model is able to discriminate well whether the customer will churn or not

Confusion Matrix

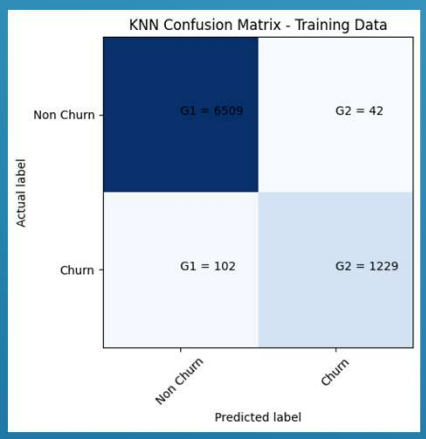

Fig 46

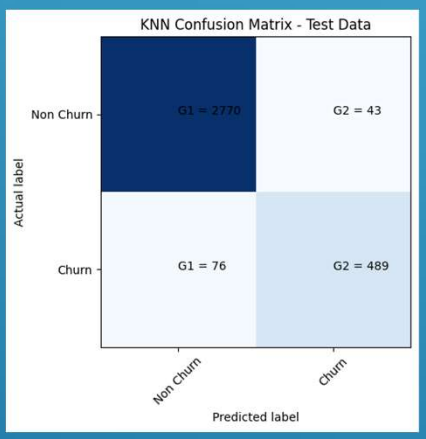

Fig 47

AUC ROC Curve

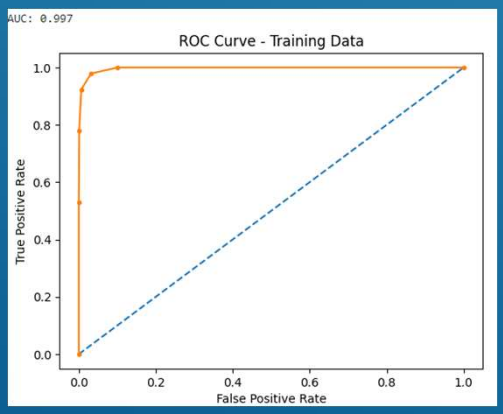

Fig 48

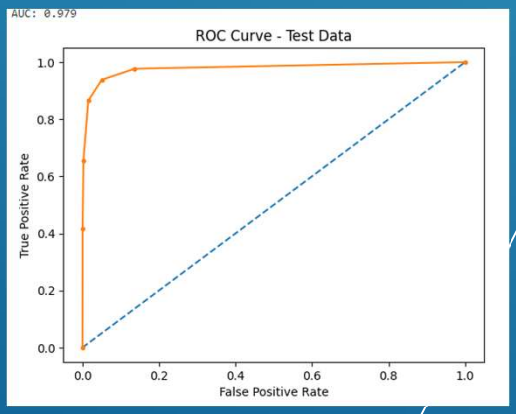

Fig 49

Model 3 – Linear Discriminant Analysis (LDA)

- Linear Discriminant Analysis (LDA) is one of the important statistical methods used for classification
- LDA aims to find a linear combination of features that best separates the classes
- It maximizes the distance between the means of the classes (between-class variance) while minimizing the spread within each class (within-class variance)

Model Performance

- Recall for churners for training and test data is 43 - 44% which is significantly lower than the recall for non-churners
- Area under the curve is close to 0.863 and 0.851 for training and test data respectively which implies the model is able to discriminate reasonably well between churners and non-churners

Confusion Matrix

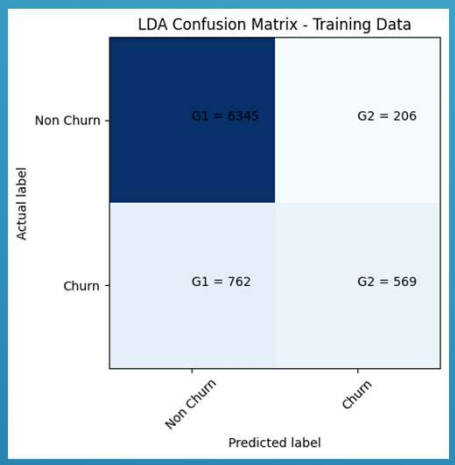

Fig 50

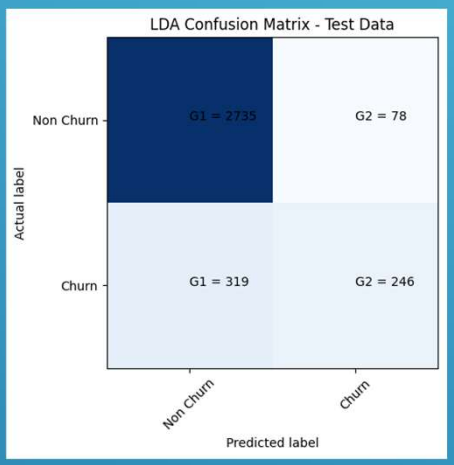

Fig 51

Coefficients and Intercepts

	Variable	Coefficient
0	tenure	-1.100656
1	city_tier	0.223081
2	cc_contacted_ly	0.218479
3	service_score	-0.121500
4	account_user_count	0.310242
5	account_segment	-0.181098
6	cc_agent_score	0.384062
7	rev_per_month	0.320831
8	complain_ly	0.846247
9	rev_growth_yoy	-0.053214
10	coupon_used_for_payment	0.142592
11	day_since_cc_connect	-0.342510
12	cashback	0.041569
13	payment_Credit Card	-0.423336
14	payment_Debit Card	-0.432376
15	payment_E wallet	-0.123547
16	payment_UPI	-0.242719
17	gender_Male	0.127541
18	marital_status_Married	-0.059391
19	marital_status_Single	0.439610
20	login_device_Mobile	-0.163534

Intercept: [-2.61335894]

Fig 52

- Like Logistic Regression model, tenure and complain last year have the most impact on whether the customer will churn or not
- Lower the tenure, higher the chance the customer will churn
- If the customer has complained last year, he's more likely to churn
- If the customer has not contacted the customer care longer, they are more likely to churn
- Single people are more likely to churn

ROC-AUC curve and Classification Report have been provided in Appendix

Model 4 – Gaussian Naïve Bayes

- Gaussian Naive Bayes is a probabilistic classifier based on applying Bayes' theorem with the "naive" assumption of conditional independence between every pair of features given the class label

- The theorem can be mathematically expressed as:

$$P(A|B)=P(B|A)\cdot P(A)/P(B)$$

- For a new observation, the Naive Bayes classifier calculates the posterior probability for each class using Bayes' theorem

- It then assigns the class with the highest posterior probability to the observation

Model Performance

- Recall for churners for training data is 57%
- Recall for churners for test data is 59%
- This suggests the model may be underfitting

Confusion Matrix

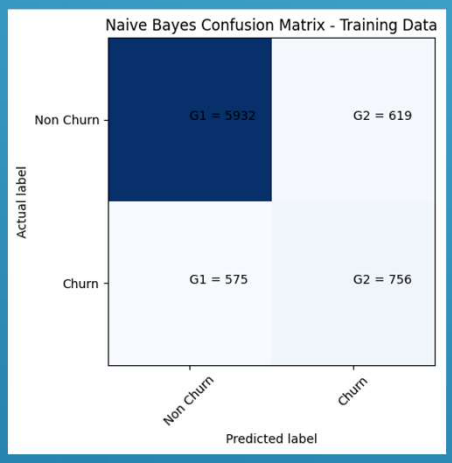

Fig 53

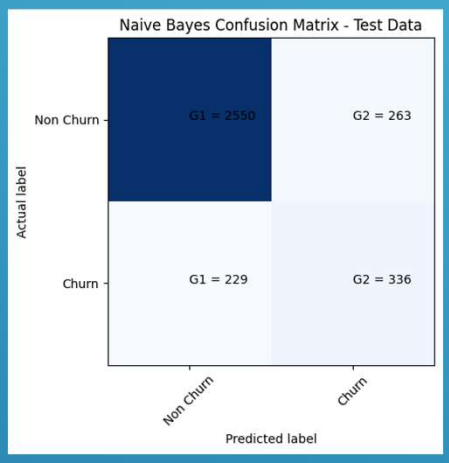

Fig 54

AUC ROC Curve

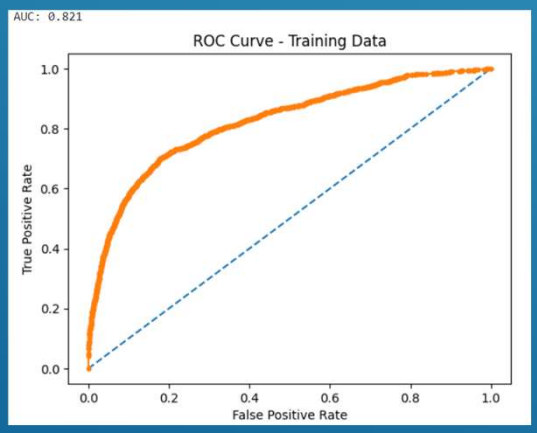

Fig 55

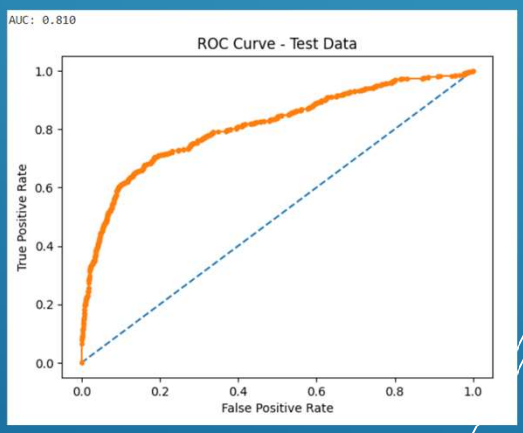

Fig 56

- Area under the curve for training data is 82.1%
- Area under the curve for test data is 81%
- The model is not performing as well as KNN

Classification Report has been provided in Appendix

Model 5 – Decision Tree

- Decision tree is a supervised learning algorithm
- It represents a model that makes decisions based on a series of questions and their answers structures in a tree like graph of decisions
- It starts with the root node and ends with the terminal node
- Entropy & Information Gain: In classification, decision trees often use entropy (a measure of impurity) and information gain (reduction in entropy) to decide where to split
- Entropy=−p 1 log 2 (p 1)−p 2 log 2 (p 2) where p1 and p2 are the proportions of the churners and non churners in the dataset
- Gini=1−(p1^2+p2^2) where p1 and p2 are the proportions of the churners and non churners in the dataset

Model Performance

- Recall for churners for training data is 100%
- Recall for churners for test data is 84%
- The recall for test data is significantly less than the training data suggesting the model is overfitting the data

Confusion Matrix

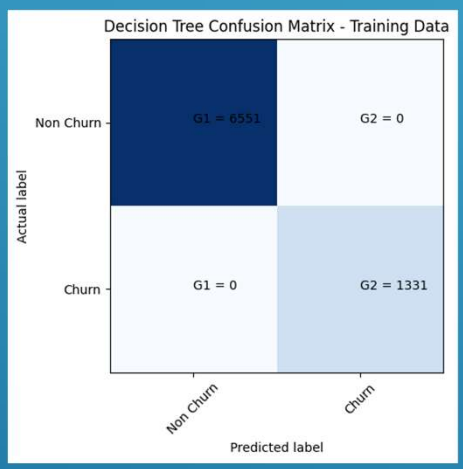

Fig 57

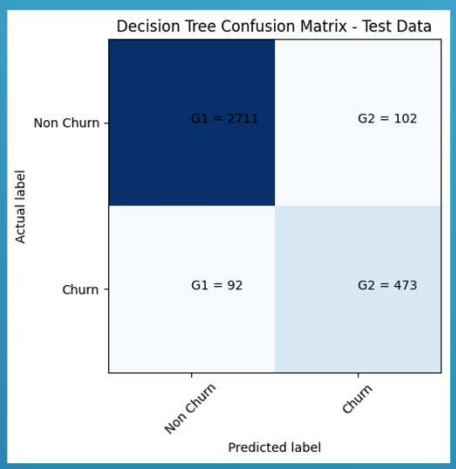

Fig 58

Model Interpretation – Feature Importance

	Imp
tenure	0.312058
day_since_cc_connect	0.083834
cashback	0.068031
rev_growth_yoy	0.058991
cc_contacted_ly	0.055615
complain_ly	0.053505
rev_per_month	0.050215
cc_agent_score	0.047432
city_tier	0.045241
marital_status_Single	0.030832
account_segment	0.029609
account_user_count	0.025419
payment_Credit Card	0.025357
login_device_Mobile	0.023770
payment_Debit Card	0.022589
service_score	0.015845
payment_E wallet	0.013929
gender_Male	0.012517
marital_status_Married	0.008500
coupon_used_for_payment	0.008358
payment_UPI	0.008354

Fig 59

- Tenure is the most important feature that plays a significant role in predicting whether a customer churn compared to other features in the dataset
- It has an importance of 0.31 which means 31% of the predictive power of the tree is attributed to this feature
- Day_since_cc_connect come in a distant second with roughly 8.38% importance

ROC-AUC curve and Classification Report have been provided in Appendix

Model Comparison – Basic Models

Training Data

	Model	Train Accuracy	Train F1 non churn	Train F1 churn	Train Recall non churn	Train Recall churn	Train Precision non churn	Train Precision churn	AUC
4	Decision Tree	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.000
1	KNN	0.98	0.99	0.94	0.99	0.92	0.98	0.97	0.997
0	Logistic Regression	0.88	0.93	0.56	0.97	0.46	0.90	0.73	0.873
2	LDA	0.88	0.93	0.54	0.97	0.43	0.89	0.73	0.863
3	Naive Bayes	0.85	0.91	0.56	0.91	0.57	0.91	0.55	0.821

Fig 60

Test Data

	Model	Test Accuracy	Test F1 non churn	Test F1 churn	Test Recall non churn	Test Recall churn	Test Precision non churn	Test Precision churn	AUC
1	KNN	0.96	0.98	0.89	0.98	0.87	0.97	0.92	0.979
4	Decision Tree	0.94	0.97	0.83	0.96	0.84	0.97	0.82	0.900
0	Logistic Regression	0.88	0.93	0.57	0.97	0.45	0.90	0.76	0.859
2	LDA	0.88	0.93	0.55	0.97	0.44	0.90	0.76	0.851
3	Naive Bayes	0.85	0.91	0.58	0.91	0.59	0.92	0.56	0.810

Fig 61

- KNN is the best performing model among all the basic models with
  - 96% accuracy
  - 98% recall for non churners
  - 87% recall for churners

- Decision Tree is the next best performer with
  - 94% accuracy
  - 96% recall for non churners
  - 84% for churners

Ensemble Modeling

Model 6 – Random Forest

- A Random Forest is an ensemble learning method which operates by constructing a multitude of decision trees during training
- Each tree in the forest is trained on a random subset of the training data, selected with replacement. This process is known as bagging and helps reduce variance and overfitting
- Each tree votes on the class and the majority vote determines the final class prediction

Model Performance

- Recall for churners for training and test data are 100% and 84% respectively
- The recall for test data is significantly less than the training data suggesting the model is overfitting the data
- Area under the curve for test data is 99%

Confusion Matrix

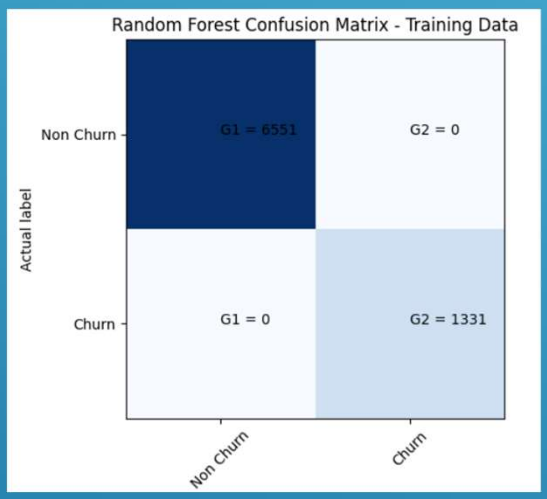

Fig 62

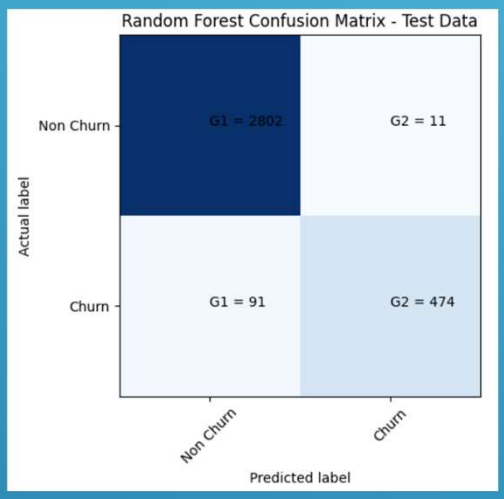

Fig 63

Model Interpretation – Feature Importance

	Imp
tenure	0.253997
cashback	0.087234
cc_contacted_ly	0.070021
day_since_cc_connect	0.068419
complain_ly	0.062587
rev_growth_yoy	0.061415
rev_per_month	0.055869
cc_agent_score	0.054799
account_segment	0.040596
account_user_count	0.036711
coupon_used_for_payment	0.028874
city_tier	0.025373
marital_status_Single	0.024287
service_score	0.021458
login_device_Mobile	0.021208
gender_Male	0.020734
payment_Credit Card	0.017896
payment_Debit Card	0.016648
marital_status_Married	0.014893
payment_E wallet	0.010381
payment_UPI	0.006598

Fig 64

- Tenure is the most important feature that plays a significant role in predicting whether a customer will churn compared to other features in the dataset just like decision tree, but the importance has reduced a bit compared to decision tree
- It has an importance of 0.25 which means 25% of the predictive power of the tree is attributed to this feature
- Cashback, cc_contacted_ly, day_since_cc_connect come in a distant second in feature importance with 7 to 8% range

ROC-AUC curve and Classification Report have been provided in Appendix

Model 7 – Bagging Classifier

- A Bagging Classifier is an ensemble machine learning algorithm that improves the stability and accuracy of a predictive model by combining multiple versions of the same type of model with random subsets of the training dataset
- For each of these subsets, a separate decision tree is trained
- Because each model is trained on a slightly different dataset, they each learn different patterns
- Final prediction is typically made by majority voting, where the class that receives the most votes from the individual models is chosen

Model Performance

- Recall for churners for training data is 100%
- Recall for churners for test data is 86%
- The recall for test data is significantly less than the training data suggesting the model is overfitting the data
- Area under the curve for test data is 98.6%
- Model is able to discriminate between the classes very well

Confusion Matrix

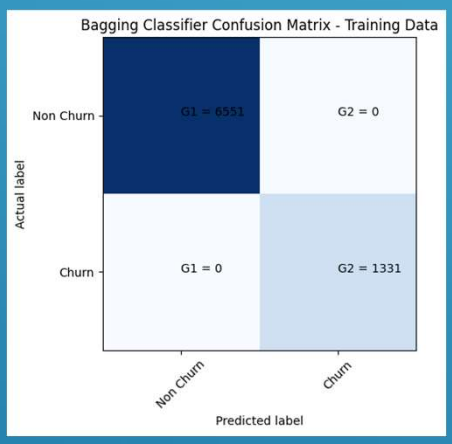

Fig 65

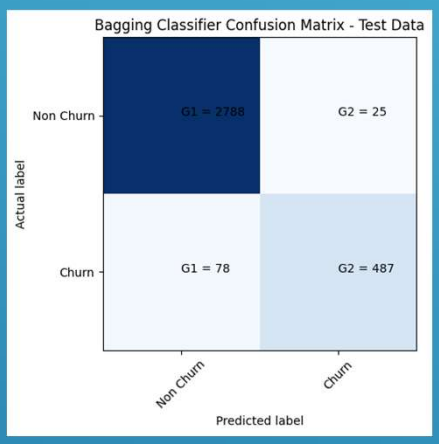

Fig 66

Model Interpretation – Feature Importance

	Imp
tenure	0.308053
day_since_cc_connect	0.074948
cashback	0.074802
cc_contacted_ly	0.066160
cc_agent_score	0.059534
rev_growth_yoy	0.058377
rev_per_month	0.054747
complain_ly	0.054021
account_segment	0.031646
city_tier	0.031419
account_user_count	0.029924
marital_status_Single	0.026053
login_device_Mobile	0.021353
coupon_used_for_payment	0.019437
payment_Credit Card	0.016705
gender_Male	0.015650
service_score	0.013966
payment_Debit Card	0.013645
payment_E wallet	0.012511
marital_status_Married	0.010906
payment_UPI	0.006145

Fig 67

- Tenure is the most important feature that plays a significant role in predicting whether a customer will churn compared to other features in the dataset just like decision tree, but the importance has reduced a bit compared to decision tree
- It has an importance of 0.308 which means 30.8% of the predictive power of the tree is attributed to this feature
- Day_since_cc_connect, cashback, cc_contacted_ly come in a distant second in feature importance with 7 to 8% range

ROC-AUC curve and Classification Report have been provided in Appendix

Model 8 – AdaBoost Classifier

- An AdaBoost Classifier is an ensemble learning algorithm that combines multiple weak learners (typically simple models like decision trees) to create a strong predictive model
- The algorithm starts by training a weak learner (e.g., a decision tree stump, which is a tree with only one split) on the original dataset
- After the first model is trained, AdaBoost assigns a weight to each training instance based on whether it was correctly or incorrectly classified
- The algorithm then increases the weightage of the misclassified instances making them more influential in the next model
- The process is repeated for a specified number of iterations or until the model's performance stops improving
- The final prediction is made by taking a weighted sum of the predictions from all the weak learners with more weightage given to better performing models

Model Performance

- Recall for churners for training data and test data is 59%
- Churn recall is very less compared to non churn recall
- This could be due to class imbalance in the data

Confusion Matrix

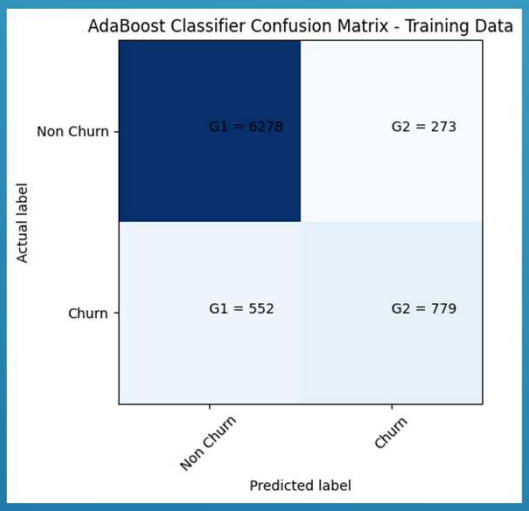

Fig 68

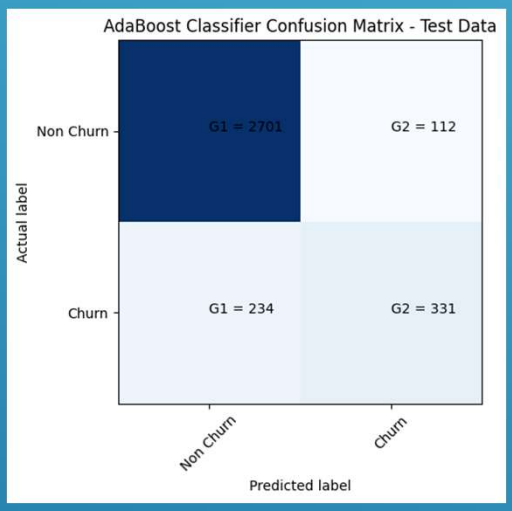

Fig 69

Model Interpretation – Feature Importance

	Imp
tenure	0.18
cashback	0.12
day_since_cc_connect	0.10
account_segment	0.08
account_user_count	0.06
rev_per_month	0.06
rev_growth_yoy	0.06
cc_contacted_ly	0.04
cc_agent_score	0.04
complain_ly	0.04
payment_E wallet	0.02
marital_status_Single	0.02
marital_status_Married	0.02
gender_Male	0.02
payment_UPI	0.02
coupon_used_for_payment	0.02
payment_Debit Card	0.02
payment_Credit Card	0.02
city_tier	0.02
service_score	0.02
login_device_Mobile	0.02

Fig 70

- Tenure, cashback and day_since_cc_connect play important roles in identifying whether the customer will churn with 18%, 12% and 10% importance

ROC-AUC curve and Classification Report have been provided in Appendix

Model 9 – Gradient Boosting Classifier

- A Gradient Boosting Classifier is an ensemble machine learning technique that builds a strong predictive model by sequentially adding weak learners to correct the errors made by previous models
- The algorithm starts by creating an initial model, typically predicting a constant value for all instances (for example, average class probability)
- The algorithm then calculates the residuals (errors) of the predictions made by the initial model. These residuals represent the difference between the actual values and the predicted values
- A new weak learner (often a small decision tree) is trained to predict these residuals, essentially trying to correct the errors made by the previous model
- The predictions from the weak learner are then added to the existing model's predictions
- This process is repeated until the model's performance ceases to improve
- After all iterations, the final model is a combination of all the weak learners, typically resulting in a strong classifier that can handle complex patterns in the data

Model Performance

- Recall for churners for training data is 65%
- Recall for churners for test data is 63%
- Churn recall is significantly lower than non churn recall
- This could be due to class imbalance in the data
- Area under the curve for training data is 95.3%
- Area under the curve for test data is 93.1%
- Model is not performing quite well in discriminating the classes

Confusion Matrix

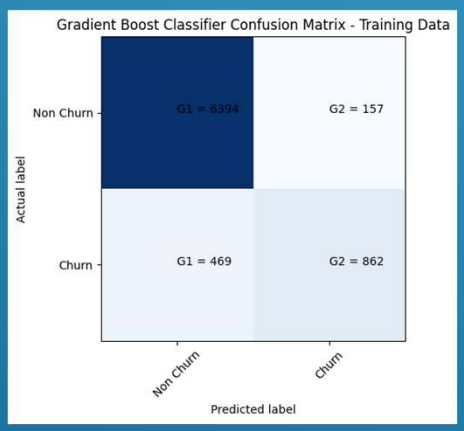

Fig 54

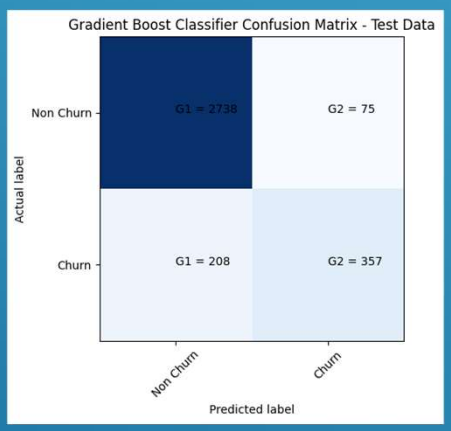

Fig 55

Model Interpretation – Feature Importance

(ROC-AUC curve and Classification Report have been provided in Appendix)

	Imp
tenure	0.551746
complain_ly	0.109375
day_since_cc_connect	0.056790
rev_per_month	0.049016
cc_agent_score	0.043128
account_segment	0.035353
marital_status_Single	0.024171
account_user_count	0.022706
cc_contacted_ly	0.020261
city_tier	0.015200
cashback	0.014042
login_device_Mobile	0.010231
rev_growth_yoy	0.009998
coupon_used_for_payment	0.009357
payment_E wallet	0.008694
payment_Credit Card	0.008447
gender_Male	0.004362
marital_status_Married	0.002971
service_score	0.002492
payment_UPI	0.000863
payment_Debit Card	0.000796

Fig 58

- Tenure is the most important feature that plays a significant role in predicting whether a customer will churn compared to other features in the dataset just like decision tree
- It has an importance of 0.55 which means 55% of the predictive power of the tree is attributed to this feature
- Complain_ly plays the next important role with 11% importance

Model 10 – XG Boosting Classifier

- XGBoost (Extreme Gradient Boosting) is an optimized and highly efficient implementation of the gradient boosting algorithm, designed for speed and performance
- Like traditional gradient boosting, XGBoost builds an ensemble of decision trees sequentially, where each new tree attempts to correct the errors made by the previous trees
- XGBoost introduces regularization terms (both L1 and L2) to the objective function to control the complexity of the model.
- This helps prevent overfitting, making XGBoost more robust than traditional gradient boosting implementations
- It minimizes a loss function (log loss for classification) using gradient descent
- XGBoost is designed to be highly efficient, with support for parallel processing
- This allows it to build trees faster by using multiple cores, making it much faster than traditional gradient boosting algorithms

Model Performance

- Recall for churners for training data is 100%
- Recall for churners for test data is 87%
- The recall for test data is significantly less than the training data suggesting the model is overfitting the data
- Area under the curve for training data is 100%
- Area under the curve for test data is 99.1%
- Model is able to discriminate between the classes very well

Confusion Matrix

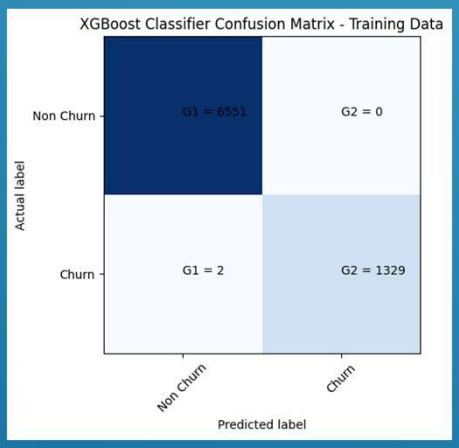

Fig 71

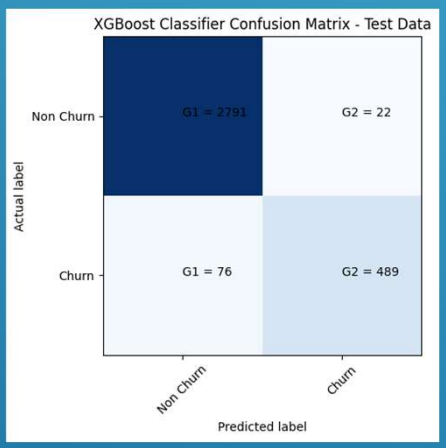

Fig 72

	Imp
tenure	0.179271
complain_ly	0.102719
account_segment	0.059142
city_tier	0.054882
marital_status_Single	0.051055
day_since_cc_connect	0.045344
cc_agent_score	0.044865
payment_Credit Card	0.044340
login_device_Mobile	0.043775
payment_E wallet	0.040730
rev_per_month	0.039034
account_user_count	0.038920
payment_UPI	0.036225
marital_status_Married	0.032810
payment_Debit Card	0.031329
gender_Male	0.030931
coupon_used_for_payment	0.030517
rev_growth_yoy	0.026922
cc_contacted_ly	0.025166
service_score	0.021195
cashback	0.020830

- Tenure is the most important feature that plays a significant role in predicting whether a customer will churn compared to other features in the dataset just like decision tree
- It has an importance of 0.17 which means 17% of the predictive power of the tree is attributed to this feature
- Complain_ly plays the next important role with 10% importance

ROC-AUC curve and Classification Report have been provided in Appendix

Model Comparison

Training Data

	Model	Train Accuracy	Train F1 non churn	Train F1 churn	Train Recall non churn	Train Recall churn	Train Precision non churn	Train Precision churn	AUC
4	Decision Tree	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.000
5	Random Forest	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.000
6	Bagging Classifier	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.000
9	XGBoost Classifier	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.000
1	KNN	0.98	0.99	0.94	0.99	0.92	0.98	0.97	0.997
8	Gradient Boosting Classifier	0.92	0.95	0.73	0.98	0.65	0.93	0.85	0.953
7	AdaBoost Classifier	0.90	0.94	0.65	0.96	0.59	0.92	0.74	0.917
0	Logistic Regression	0.88	0.93	0.56	0.97	0.46	0.90	0.73	0.873
2	LDA	0.88	0.93	0.54	0.97	0.43	0.89	0.73	0.863
3	Naive Bayes	0.85	0.91	0.56	0.91	0.57	0.91	0.55	0.821

Fig 73

Test Data

	Model	Test Accuracy	Test F1 non churn	Test F1 churn	Test Recall non churn	Test Recall churn	Test Precision non churn	Test Precision churn	AUC
5	Random Forest	0.97	0.98	0.90	1.00	0.84	0.97	0.98	0.990
6	Bagging Classifier	0.97	0.98	0.90	0.99	0.86	0.97	0.95	0.986
9	XGBoost Classifier	0.97	0.98	0.91	0.99	0.87	0.97	0.96	0.991
1	KNN	0.96	0.98	0.89	0.98	0.87	0.97	0.92	0.979
4	Decision Tree	0.94	0.97	0.83	0.96	0.84	0.97	0.82	0.900
8	Gradient Boosting Classifier	0.92	0.95	0.72	0.97	0.63	0.93	0.83	0.931
7	AdaBoost Classifier	0.90	0.94	0.66	0.96	0.59	0.92	0.75	0.905
0	Logistic Regression	0.88	0.93	0.57	0.97	0.45	0.90	0.76	0.859
2	LDA	0.88	0.93	0.55	0.97	0.44	0.90	0.76	0.851
3	Naive Bayes	0.85	0.91	0.58	0.91	0.59	0.92	0.56	0.810

Fig 74

- Random forest, Bagging Classifier, XGBoost Classifier, KNN and Decision Tree models have performed very well in all the metrics

Model Tuning – Logistic Regression

- Variance Inflation Factor (VIF) was checked to see if there's any multicollinearity affecting the model performance
- Logistic Regression model was also tuned using RFE (Recursive Feature Elimination)

	Features	VIF
0	tenure	1.282526
1	city_tier	1.402313
2	cc_contacted_ly	1.019630
3	service_score	1.209550
4	account_user_count	1.146011
5	account_segment	1.248245
6	cc_agent_score	1.015272
7	rev_per_month	1.089170
8	complain_ly	1.009761
9	rev_growth_yoy	1.021443
10	coupon_used_for_payment	1.282429
11	day_since_cc_connect	1.287499
12	cashback	1.600285
13	payment_Credit Card	3.043221
14	payment_Debit Card	3.239589
15	payment_E wallet	2.300608
16	payment_UPI	1.673332
17	gender_Male	1.013089
18	marital_status_Married	2.148889
19	marital_status_Single	2.160165
20	login_device_Mobile	1.009844

Fig 75

VIF values are less than 5, so multicollinearity is not affecting the model performance

Important features returned by RFE:

	Feature	Rank
0	tenure	1
17	gender_Male	1
16	payment_UPI	1
14	payment_Debit Card	1
13	payment_Credit Card	1
11	day_since_cc_connect	1
19	marital_status_Single	1
8	complain_ly	1
10	coupon_used_for_payment	1
6	cc_agent_score	1
4	account_user_count	1
2	cc_contacted_ly	1
1	city_tier	1
7	rev_per_month	1
20	login_device_Mobile	1

Fig 76

Model Performance after RFE tuning:

```
Training Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.97      0.93      6551
           1       0.74      0.45      0.56      1331

    accuracy                           0.88      7882
   macro avg       0.82      0.71      0.75      7882
weighted avg       0.87      0.88      0.87      7882
```

Fig 77

Model performance has not shown much improvement after RFE

```
Test Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.97      0.93      2813
           1       0.77      0.45      0.57       565

    accuracy                           0.88      3378
   macro avg       0.83      0.71      0.75      3378
weighted avg       0.88      0.88      0.87      3378
```

Fig 78

Using SMOTE

- SMOTE is Synthetic Minority Oversampling Technique is used to address the class imbalance issue in the dataset
- Minority class is churn =1
- Synthetic samples are generated along the line segments connecting minority class instances
- After using SMOTE, the ration of non-churn vs churn data is 1:0.75
- Model Score after using SMOTE:

```
Training score with SMOTE:  0.8107990230286113
Test score with SMOTE:  0.8090586145648313
```

Fig 79

Model Tuning – KNN

- Using GridSearchCV, the best parameters were found for KNN
- Best Parameters:

```
Best parameters: {'metric': 'manhattan', 'n_neighbors': 3, 'p': 1, 'weights': 'distance'}
Test set accuracy: 0.9843102427471877
```

Fig 80

- Model Performance with the above parameters:

```
Training Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      6551
           1       1.00      1.00      1.00      1331

    accuracy                           1.00      7882
   macro avg       1.00      1.00      1.00      7882
weighted avg       1.00      1.00      1.00      7882

Test Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      2813
           1       0.95      0.96      0.95       565

    accuracy                           0.98      3378
   macro avg       0.97      0.98      0.97      3378
weighted avg       0.98      0.98      0.98      3378
```

Fig 81

- Recall for non-churn and churn for training data is 100%
- Recall for non-churn and churn for test data is 99% and 96% respectively

AUC ROC Curve

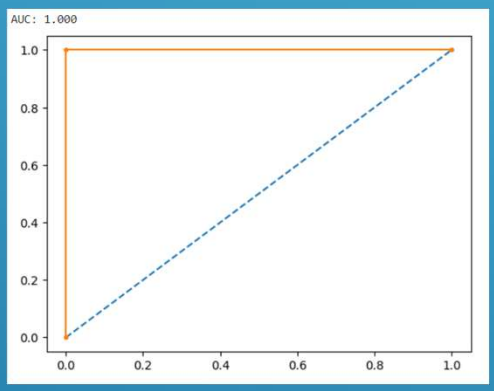

Fig 82

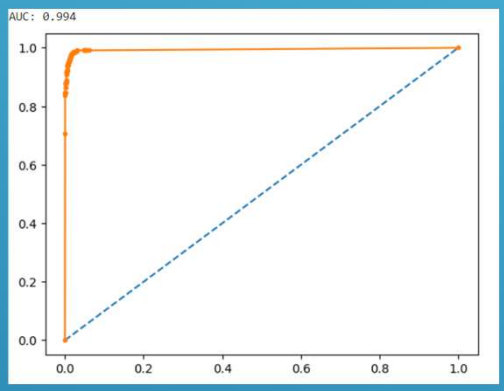

Fig 83

- Area under the curve for training data is 100%
- Area under the curve for test data is 99.4%
- Model is performing extremely well after tuning

Model Tuning – Decision Tree

- Using GridSearchCV, the best parameters were found for Decision Tree
- Best Parameters:

```
Best parameters: {'criterion': 'entropy', 'max_depth': 20, 'max_features': None, 'min_samples_leaf': 1, 'min_samples_split': 2}
```

Fig 78

- Model Performance with the above parameters:

```
Training Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      6551
           1       1.00      1.00      1.00      1331

    accuracy                           1.00      7882
   macro avg       1.00      1.00      1.00      7882
weighted avg       1.00      1.00      1.00      7882

Test Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.97      0.97      2813
           1       0.85      0.84      0.85       565

    accuracy                           0.95      3378
   macro avg       0.91      0.91      0.91      3378
weighted avg       0.95      0.95      0.95      3378
```

Fig 84

- Recall for non-churn and churn for training data is 100%
- Recall for non-churn and churn for test data is 97% and 84% respectively
- The model is still overfitting the training data

Model Tuning – Random Forest

- Using GridSearchCV, the best parameters were found for Decision Tree
- Best Parameters:

```
Best parameters: {'bootstrap': False, 'max_depth': 20, 'max_features': 'log2', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}
```

Fig 85

- Model Performance with the above parameters:

```
Training Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      6551
           1       1.00      1.00      1.00      1331

    accuracy                           1.00      7882
   macro avg       1.00      1.00      1.00      7882
weighted avg       1.00      1.00      1.00      7882

Test Classification Report:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99      2813
           1       0.98      0.89      0.93       565

    accuracy                           0.98      3378
   macro avg       0.98      0.94      0.96      3378
weighted avg       0.98      0.98      0.98      3378
```

Fig 86

- Recall for non-churn and churn for training data is 100%
- Recall for non-churn and churn for test data is 100% and 89% respectively
- The model is still overfitting the training data

- Random Forest and KNN models were also built
  - After treating the class imbalance with SMOTE
  - Without treating the outliers in the dataset
  - Performance of all the models have been summarized in Appendix

Performance of the Best 5 Models – Test Data

No	Model Name	Recall	Precision	F1 Score	Accuracy	AUC
1	KNN Tuned	0 – 99% 1 – 96%	0 – 99% 1 – 95%	0 – 99% 1 – 95%	98%	99.4%
2	KNN Tuned With SMOTE	0 – 98% 1 – 98%	0 – 100% 1 – 90%	0 – 99% 1 – 94%	98%	99.2%
3	KNN Tuned with outlier	0 – 99% 1 – 93%	0 – 99% 1 – 94%	0 – 99% 1 – 94%	98%	98.6%
4	Random Forest Tuned	0 – 100% 1 – 89%	0 – 98% 1 – 98%	0 – 99% 1 – 93%	98%	99.6%
5	Random Forest Tuned with Outliers	0 – 100% 1 – 89%	0 – 98% 1 – 98%	0 – 99% 1 – 93%	98%	99.5%

Table 3

Model Selection

- Looking at the above table, we can conclude that **KNN Tuned is the best performing of all the models**
- Even without using SMOTE, the model can predict 96% of the customers who will churn which is higher than the next best performing model (Random Forest Tuned)
- The F1 Score which is the harmonic mean of Precision and Recall is 99% and 95% for non churner and churner prediction respectively

Advantages

- KNN is one of the simplest machine learning algorithms, making it easy to understand and implement
- It doesn't require any assumptions about the underlying data distribution
- Instantaneous learning and adaptability

Disadvantages of KNN

- KNN is computationally expensive, especially with large datasets as it needs to store and search through the entire training set
- It is quite memory intensive, will be slow for large datasets
- It is less interpretable
- The algorithm does not distinguish between noisy and reliable instances, so noise can directly influence the majority voting process
- KNN has no explicit training phase, so all the computational burden is on the prediction phase

Please Note: Performance of all the models have been provided in the Appendix Page 14

Model Selection

Second Best Model

- **Random Forest Tuned (with or without outlier treatment) is the next best performing model**
- It seems to suffer from overfitting issues because the recall for churners in test data is 89%, 11% less than that of training data
- It is computationally less intensive during prediction phase
- Further hyper parameter tuning can be done to improve its performance

Advantages

- Random Forest provides insights into feature importance
- It can also handle noisy data and outliers

Disadvantages

- It has a risk of overfitting
- Memory intensive as it creates multiple decision trees

Optimum Model and Business Implications

- **KNN model can be used if the company can handle its computational requirements**
- **Otherwise, Random Forest can be used for predicting whether the customer will churn**

Best Performing Model

KNN Tuned – Confusion Matrix

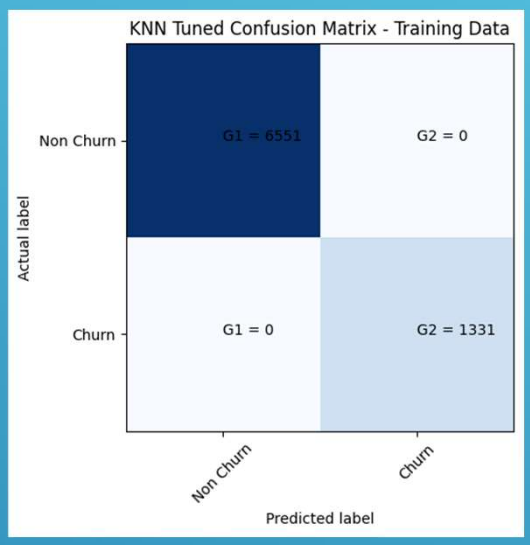

Fig 87

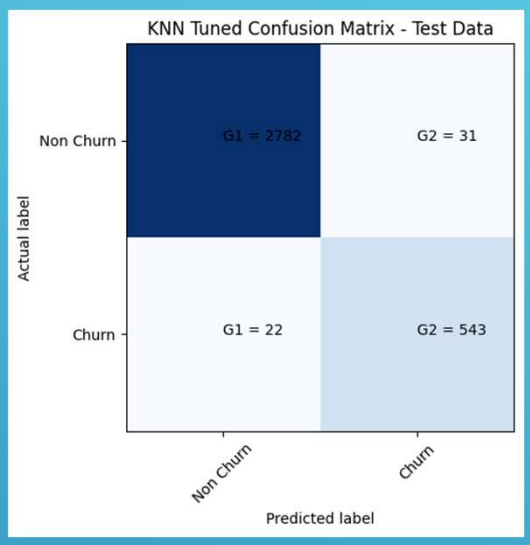

Fig 88

- In the training data all the churners and non churners have been predicted correctly by the tuned KNN model
- In the test data
  - 2782 non churners have been identified correctly by the model
  - 543 churners have been identified correctly by the model
  - 22 churners have been misclassified as non-churners
  - 31 non churners have been misclassified as churners

KNN Tuned – AUC – ROC Curve

Training Data

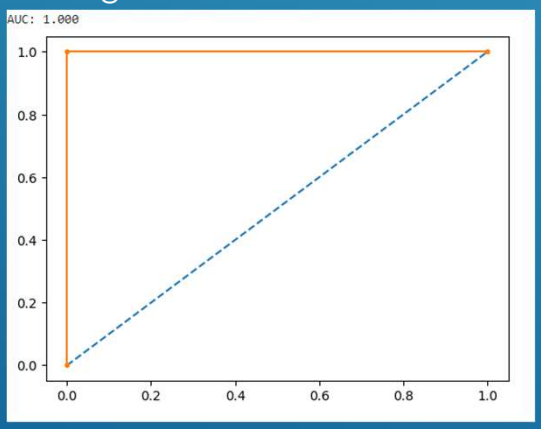

Fig 88

Test Data

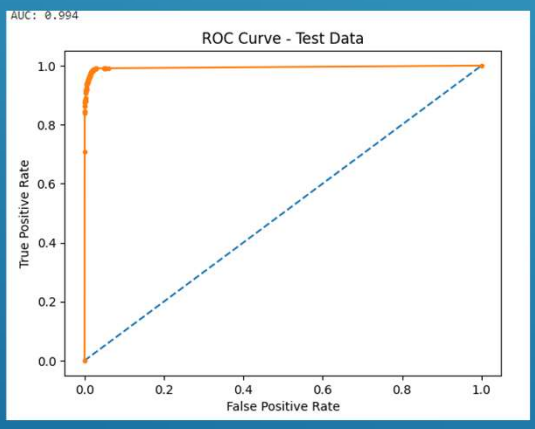

Fig 89

The model is able to distinguish very well between classes with the AUC score of 99.4%

Second Best Performing Model

Random Forest Tuned – Confusion Matrix

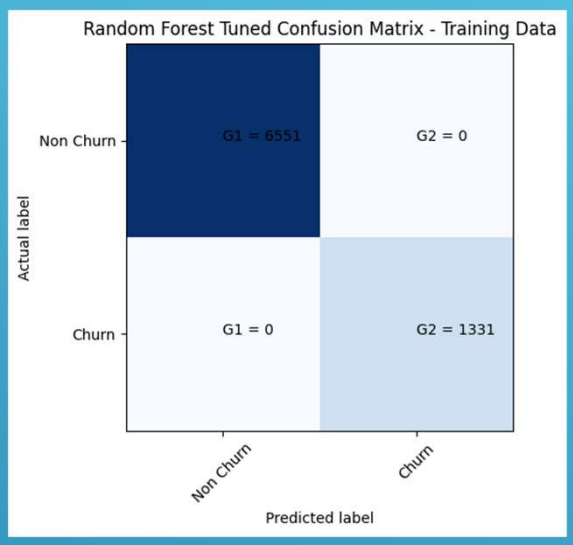

Fig 89

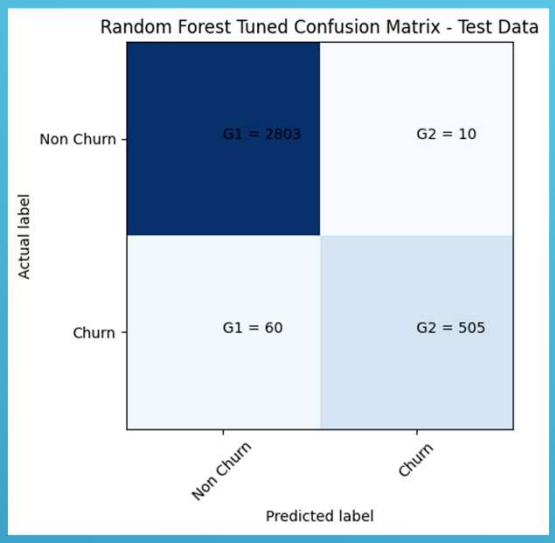

Fig 90

- In the training data, all the churners and non churners have been predicted correctly by the tuned Random Forest model
- In the test data
  - 2803 non churners have been identified correctly by the model
  - 505 churners have been identified correctly by the model
  - 60 churners have been misclassified as non-churners
  - 10 non churners have been misclassified as churners

Random Forest Tuned – AUC – ROC Curve

Training Data

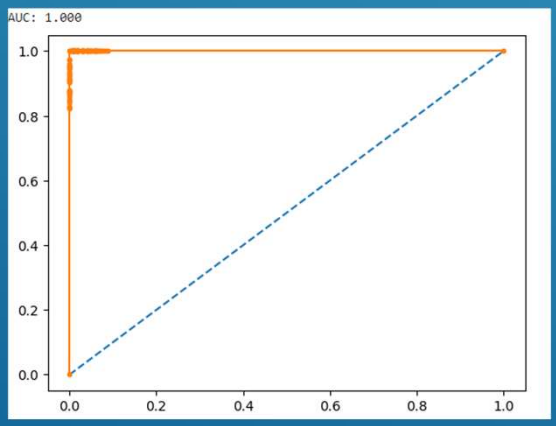

Fig 91

Test Data

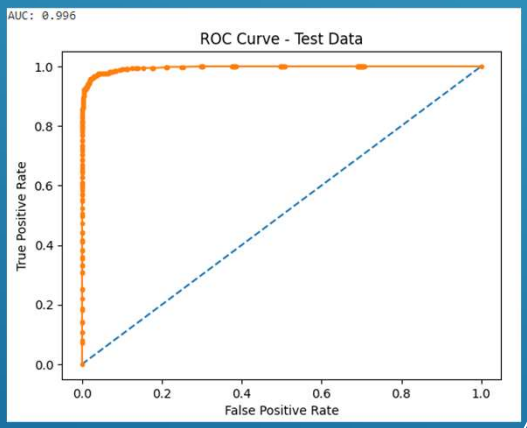

Fig 92

The model can distinguish very well between classes with the AUC score of 99.6%

Actionable Insights from the Models

- Though the other models did not perform as well as KNN and Random Forest, they have provided important insights about the independent variables
- **Tenure is the most important factor in predicting the churn in all the models**
- **Customers with longer tenure tend to churn less**
- Other variables that contribute significantly to the churn are:
  - **Complain last year** – customers who have complained the previous year have churned more
  - **Marital Status** – single people are more likely to churn
  - **Gender** - Males are more likely to churn, so marketing campaign to attract more new female customers can be devised
  - Cashback has also played a minor role in churn prediction

Recommendations

- Loyalty programs can be offered to retain the customers to increase their tenure
- Cashback on purchases can be increased by 1% to improve customer retention
- Detailed feedback can be collected to analyze what drives customer satisfaction and dissatisfaction
- Customer service personnel can be given regular training to improve customer service
- Targeted marketing for city tier 3 customers based on their purchase pattern
- Market share in city tier 2 and 3 can be increased by targeted marketing
- More focus can be given for the accounts with high number of users by giving rewards and discounts
- Customers in cluster 3 (where the churn rate is high) can be reached out with special offers
- Constant engagement with the customers through email and social media platforms
- Continuous improvement of products and services
- Inactive customers can be reached out with personalized messages and special offers like discounts on birthdays and anniversaries to win them back
- More focus needs to given to regular plus customers as they have churned the most, they can be upgraded to Super category with special offers
- Single customers can be given special offers on the products they buy more to reduce the churn rate
- Super plus and HNI customers market share can be increased as they spend more and churn less

Thank You!